Department of Education & Workforce

Ohio's State Tests

ITEM RELEASE

SPRING 2025

GEOMETRY

Table of Contents

Content Summary and Answer Key iii

Depth of Knowledge (DOK) ix

Question 1: Question and Scoring Guidelines 1
Question 1: Sample Response 4

Question 3: Question and Scoring Guidelines 5
Question 3: Sample Response 7

Question 5: Question and Scoring Guidelines 8
Question 5: Sample Responses 11

Question 11: Question and Scoring Guidelines 18
Question 11: Sample Responses 21

Question 12: Question and Scoring Guidelines 28
Question 12: Sample Responses 31

Question 20: Question and Scoring Guidelines 40
Question 20: Sample Response 43

Question 21: Question and Scoring Guidelines 44
Question 21: Sample Responses 47

Question 22: Question and Scoring Guidelines 51
Question 22: Sample Response 54

Question 24: Question and Scoring Guidelines 55
Question 24: Sample Response 57

Question 25: Question and Scoring Guidelines 58
Question 25: Sample Response 61

Question 27: Question and Scoring Guidelines 62
Question 27: Sample Responses 65

Question 30: Question and Scoring Guidelines 69
Question 30: Sample Responses 73

Question 33: Question and Scoring Guidelines..... 86
Question 33: Sample Response 88

Question 35: Question and Scoring Guidelines..... 89
Question 35: Sample Responses..... 92

Question 37: Question and Scoring Guidelines..... 98
Question 37: Sample Responses..... 101

Question 41: Question and Scoring Guidelines..... 105
Question 41: Sample Responses..... 109

Question 42: Question and Scoring Guidelines..... 116
Question 42: Sample Responses..... 119

Question 43: Question and Scoring Guidelines..... 123
Question 43: Sample Response 126

Question 45: Question and Scoring Guidelines..... 127
Question 45: Sample Response 129

Question 46: Question and Scoring Guidelines..... 130
Question 46: Sample Responses..... 133

Question 51: Question and Scoring Guidelines..... 140
Question 51: Sample Response 142

Geometry
Spring 2025 Item Release
Content Summary and Answer Key

Question No.*	Item Type	Content Cluster	Content Standard	Depth of Knowledge	Answer Key	Points
1	Multiple Choice Item	Understand the relationships between lengths, area, and volumes.	Understand how and when changes to the measures of a figure (lengths or angles) result in similar and non-similar figures. *(G.GMD.5)*	Level 1	C	1 point
3	Multiple Choice Item	Understand indepen-dence and conditional probability, and use them to interpret data.	Recognize and explain the concepts of conditional probability and independence in everyday language and everyday situations. *For example, compare the chance of having lung cancer if you are a smoker with the chance of being a smoker if you have lung cancer. (S.CP.5)*	Level 1	D	1 point
5	Equation Item	Prove and apply theorems both formally and informally involving similarity using a variety of methods.	Use congruence and similarity criteria for triangles to solve problems and to justify relationships in geometric figures that can be decomposed into triangles. *(G.SRT.5)*	Level 3	---	1 point
11	Equation Item	Experiment with transform-ations in the plane.	Represent transformations in the plane using, e.g., transparencies and geometry software; describe transformations as functions that take points in the plane as inputs and give other points as outputs. Compare transformations that preserve distance and angle to those that do not (e.g., translation versus horizontal stretch). *(G.CO.2)*	Level 1	---	1 point

** The question numbers denote each question's place in the online test. Items are numbered sequentially on the practice site.*

Geometry
Spring 2025 Item Release
Content Summary and Answer Key

Question No.*	Item Type	Content Cluster	Content Standard	Depth of Knowledge	Answer Key	Points
12	Equation Item	Apply geometric concepts in modeling situations.	Apply geometric methods to solve design problems (e.g., designing an object or structure to satisfy physical constraints or minimize cost; working with typographic grid systems based on ratios). *(G.MG.3)*	Level 2	---	1 point
20	Multiple Choice Item	Understand similarity in terms of similarity transform-ations.	Given two figures, use the definition of similarity in terms of similarity transformations to decide if they are similar; explain using similarity transformations the meaning of similarity for triangles as the equality of all corresponding pairs of angles and the proportionality of all corresponding pairs of sides. *(G.SRT.2)*	Level 2	D	1 point
21	Equation Item	Understand indepen-dence and conditional probability, and use them to interpret data.	Understand that two events A and B are independent if and only if the probability of A and B occurring together is the product of their probabilities, and use this characterization to determine if they are independent. *(S.CP.2)*	Level 1	---	1 point
22	Multiple Choice Item	Prove geometric theorems both formally and informally using a variety of methods.	Prove and apply theorems about lines and angles. Theorems include but are not restricted to the following: vertical angles are congruent; when a transversal crosses parallel lines, alternate interior angles are congruent and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints. *(G.CO.9)*	Level 2	A	1 point

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Geometry
Spring 2025 Item Release
Content Summary and Answer Key

Question No.*	Item Type	Content Cluster	Content Standard	Depth of Knowledge	Answer Key	Points
24	Multiple Choice Item	Classify and analyze geometric figures.	Classify two-dimensional figures in a hierarchy based on properties. *(G.CO.14)*	Level 2	D	1 point
25	Multiple Choice Item	Use the rules of probability to compute probabilities of compound events in a uniform probability model.	Apply the Addition Rule, P(A or B) = P(A) + P(B) − P(A and B), and interpret the answer in terms of the model. *(S.CP.7)*	Level 1	B	1 point
27	Inline Choice Item	Understand indepen-dence and conditional probability, and use them to interpret data.	Recognize and explain the concepts of conditional probability and independence in everyday language and everyday situations. *For example, compare the chance of having lung cancer if you are a smoker with the chance of being a smoker if you have lung cancer. (S.CP.5)*	Level 2	---	1 point

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Geometry
Spring 2025 Item Release
Content Summary and Answer Key

Question No.*	Item Type	Content Cluster	Content Standard	Depth of Knowledge	Answer Key	Points
30	Equation Item	Understand indepen-dence and conditional probability, and use them to interpret data.	Construct and interpret two-way frequency tables of data when two categories are associated with each object being classified. Use the two-way table as a sample space to decide if events are independent and to approximate conditional probabilities. *For example, collect data from a random sample of students in your school on their favorite subject among math, science, and English. Estimate the probability that a randomly selected student from your school will favor science given that the student is in tenth grade. Do the same for other subjects and compare the results. (S.CP.4)*	Level 3	---	2 points
33	Multiple Choice Item	Use the rules of probability to compute probabilities of compound events in a uniform probability model.	Find the conditional probability of A given B as the fraction of B's outcomes that also belong to A, and interpret the answer in terms of the model. *(S.CP.6)*	Level 1	A	1 point
35	Equation Item	Define trigonometric ratios and solve problems involving right triangles.	Explain and use the relationship between the sine and cosine of complementary angles. *(G.SRT.7)*	Level 1	---	1 point

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Geometry
Spring 2025 Item Release
Content Summary and Answer Key

Question No.*	Item Type	Content Cluster	Content Standard	Depth of Knowledge	Answer Key	Points
37	Equation Item	Apply geometric concepts in modeling situations.	Apply concepts of density based on area and volume in modeling situations (e.g., persons per square mile, BTUs per cubic foot). *(G.MG.2)*	Level 2	---	1 point
41	Multi-Interaction Item	Use coordinates to prove simple geometric theorems algebraically and to verify specific geometric statements.	Justify the slope criteria for parallel and perpendicular lines, and use them to solve geometric problems, e.g., find the equation of a line parallel or perpendicular to a given line that passes through a given point. *(G.GPE.5)*	Level 2	---	1 point
42	Equation Item	Understand indepen-dence and conditional probability, and use them to interpret data.	Understand that two events A and B are independent if and only if the probability of A and B occurring together is the product of their probabilities, and use this characterization to determine if they are independent. *(S.CP.2)*	Level 1	---	1 point
43	Multi-Select Item	Experiment with transform-ations in the plane.	Given a geometric figure and a rotation, reflection, or translation, draw the transformed figure using, e.g., graph paper, tracing paper, or geometry software. Specify a sequence of transformations that will carry a given figure onto another. *(G.CO.5)*	Level 2	B, D	1 point
45	Multiple Choice Item	Experiment with transform-ations in the plane.	Develop definitions of rotations, reflections, and translations in terms of angles, circles, perpendicular lines, parallel lines, and line segments. *(G.CO.4)*	Level 2	D	1 point

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Geometry
Spring 2025 Item Release
Content Summary and Answer Key

Question No.*	Item Type	Content Cluster	Content Standard	Depth of Knowledge	Answer Key	Points
46	Equation Item	Use coordinates to prove simple geometric theorems algebraically and to verify specific geometric statements.	Use coordinates to compute perimeters of polygons and areas of triangles and rectangles, e.g., using the distance formula. *(G.GPE.7)*	Level 2	---	1 point
51	Multiple Choice Item	Experiment with transform-ations in the plane.	Know precise definitions of ray, angle, circle, perpendicular line, parallel line, and line segment, based on the undefined notions of point, line, distance along a line, and arc length. *(G.CO.1)*	Level 1	D	1 point

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Depth of Knowledge (DOK)

DOK refers to the complexity of thinking required to complete a task in a given item. Items with a DOK 1 designation focus on the recall of information, such as definitions and terms, and simple procedures. Items with a DOK 2 designation require students to make decisions, solve routine problems, perform calculations, or recognize patterns. Items with a DOK 3 designation feature higher-order cognitive tasks. These DOK 3 tasks include but are not limited to: critiquing a statement and forming a conclusion; explaining, justifying, or proving a statement; or approaching abstract, complex, open-ended, and non-routine problems. Each grade's blueprint contains information about the number of points of opportunity students will encounter at each DOK level.

Table 1: *Math Descriptors* – Applying Depth of Knowledge Levels for Mathematics (Webb, 2002) & NAEP 2002 Mathematics Levels of Complexity
(M. Petit, Center for Assessment 2003, K. Hess, Center for Assessment, updated 2006)

Level 1 Recall	Level 2 Skills/Concepts	Level 3 Strategic Thinking	Level 4* Extended Thinking
a. Recall, observe, or recognize a fact, definition, term, or property b. Apply/compute a well-known algorithm (e.g., sum, quotient) c. Apply a formula d. Determine the area or perimeter of rectangles or triangles given a drawing and labels e. Identify a plane or three-dimensional figure f. Measure g. Perform a specified or routine procedure (e.g., apply rules for rounding) h. Evaluate an expression i. Solve a one-step word problem j. Retrieve information from a table or graph	a. Classify plane and three-dimensional figures b. Interpret information from a simple graph c. Use models to represent mathematical concepts d. Solve a routine problem requiring multiple steps/ decision points, or the application of multiple concepts e. Compare and/or contrast figures or statements f. Construct 2-dimensional patterns for 3-dimensional models, such as cylinders and cones g. Provide justifications for steps in a solution process h. Extend a pattern	a. Interpret information from a complex graph b. Explain thinking when more than one response is possible c. Make and/or justify conjectures d. Use evidence to develop logical arguments for a concept e. Use concepts to solve non-routine problems f. Perform a procedure with multiple steps and multiple decision points g. Generalize a pattern h. Describe, compare, and contrast solution methods i. Formulate a mathematical model for a complex situation j. Provide mathematical justifications	a. Relate mathematical concepts to other content areas b. Relate mathematical concepts to real-world applications in new situations c. Apply a mathematical model to illuminate a problem, situation d. Conduct a project that specifies a problem, identifies solution paths, solves the problem, and reports results e. Design a mathematical model to inform and solve a practical or abstract situation f. Develop generalizations of the results obtained and the strategies used and apply them to new problem situations

Table 1 *continued on next page.*

Level 1 Recall	Level 2 Skills/Concepts	Level 3 Strategic Thinking	Level 4* Extended Thinking
k. Recall, identify, or make conversions between and among representations or numbers (fractions, decimals, and percents), or within and between customary and metric measures l. Locate numbers on a number line, or points on a coordinate grid m. Solve linear equations n. Represent math relationships in words, pictures, or symbols o. Read, write, and compare decimals in scientific notation	i. Retrieve information from a table, graph, or figure and use it to solve a problem requiring multiple steps j. Translate between tables, graphs, words and symbolic notation k. Make direct translations between problem situations and symbolic notation l. Select a procedure according to criteria and perform it m. Specify and explain relationships between facts, terms, properties, or operations n. Compare, classify, organize, estimate, or order data	k. Solve a multiple-step problem and provide support with a mathematical explanation that justifies the answer l. Solve 2-step linear equations/ inequalities in one variable over the rational numbers, interpret solution(s) in the original context, and verify reasonableness of results m. Translate between a problem situation and symbolic notation that is not a direct translation n. Formulate an original problem, given a situation o. Analyze the similarities and differences between procedures p. Draw conclusion from observations or data, citing evidence	g. Apply one approach among many to solve problems h. Apply understanding in a novel way, providing an argument/ justification for the application ***NOTE:*** *Level 4 involves such things as complex restructuring of data or establishing and evaluating criteria to solve problems.*

Note: Ohio's State Tests only assess and measure DOK Levels 1 – 3 in grades K – 12. Level 4 is included in this table for informational purposes only.

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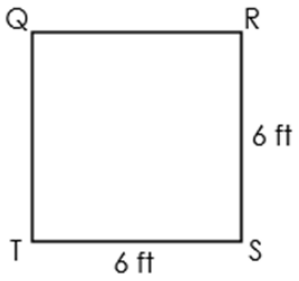
Geometry
Spring 2025 Item Release

Question 1

Question and Scoring Guidelines

Question 1

Square QRST is shown, with dimensions in feet (ft).



Felix divides each side length of QRST by 4 to create square WXYZ.

Which statement describes whether the squares QRST and WXYZ are similar?

Ⓐ no, because the squares are different sizes

Ⓑ yes, because the squares are the same size

Ⓒ yes, because corresponding side lengths are proportional

Ⓓ no, because corresponding side lengths are not proportional

Points Possible: 1

Content Cluster: Understand the relationships between lengths, area, and volumes.

Content Standard: Understand how and when changes to the measures of a figure (lengths or angles) result in similar and non-similar figures. *(G.GMD.5)*

Depth of Knowledge: Level 1

Modeling and Reasoning: No

Scoring Guidelines

Rationale for Option A: This is incorrect. The student may confuse the meaning of similarity with congruence.

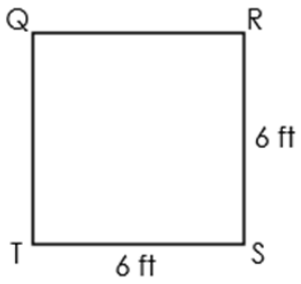
Rationale for Option B: This is incorrect. The student may think that the square remains the same size after the dilation.

Rationale for Option C: **Key** – The student understands that the squares are similar because the side lengths are scaled by the same factor, and therefore the side lengths are proportional.

Rationale for Option D: This is incorrect. The student may not understand that a dilation scales the side lengths of a figure proportionally and may think a dilation will not result in two similar squares.

Sample Response: 1 point

Square QRST is shown, with dimensions in feet (ft).



Felix divides each side length of QRST by 4 to create square WXYZ.

Which statement describes whether the squares QRST and WXYZ are similar?

- (A) no, because the squares are different sizes
- (B) yes, because the squares are the same size
- ● yes, because corresponding side lengths are proportional
- (D) no, because corresponding side lengths are not proportional

Geometry
Spring 2025 Item Release

Question 3

Question and Scoring Guidelines

Question 3

Which pair of compound events is dependent?

Ⓐ rolling a cube numbered 1 through 6, twice

Ⓑ spinning the arrow on a spinner with 6 unequal sections, twice

Ⓒ drawing two cards from a deck one at a time, with replacement

Ⓓ picking two crayons from a box one at a time, without replacement

Points Possible: 1

Content Cluster: Understand independence and conditional probability, and use them to interpret data.

Content Standard: Recognize and explain the concepts of conditional probability and independence in everyday language and everyday situations. *For example, compare the chance of having lung cancer if you are a smoker with the chance of being a smoker if you have lung cancer. (S.CP.5)*

Depth of Knowledge: Level 1

Modeling and Reasoning: Yes

Scoring Guidelines

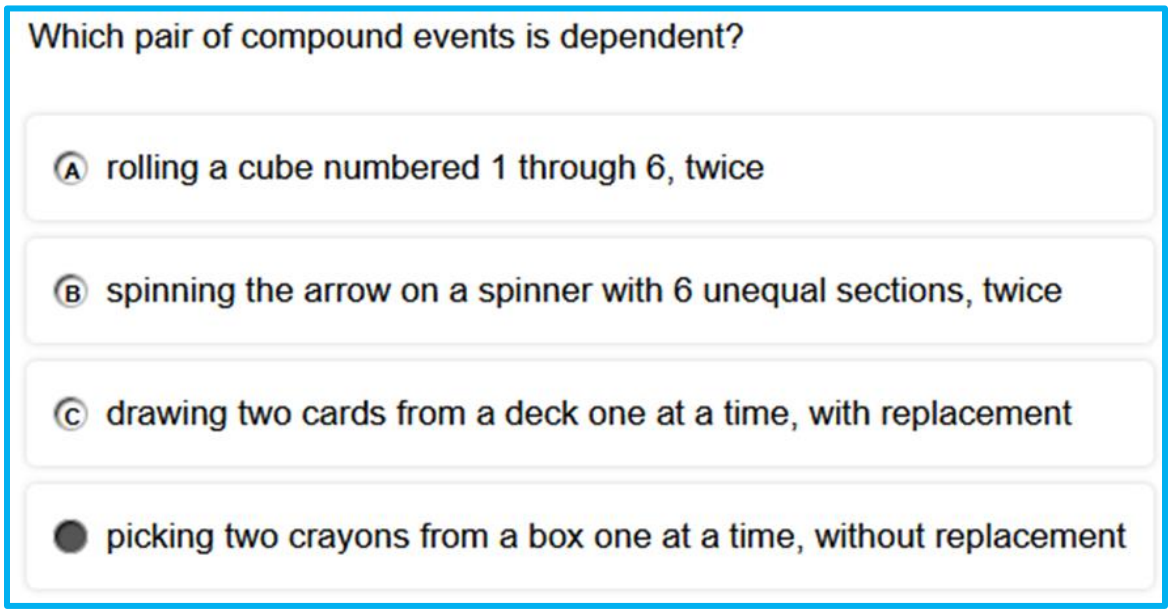
Rationale for Option A: This is incorrect. The student may think that once one number is rolled, it cannot be rolled again the second time.

Rationale for Option B: This is incorrect. The student may think the sections not being equally sized affects the probability when spinning the second time.

Rationale for Option C: This is incorrect. The student may confuse "dependent" and "independent," thinking events with replacement are dependent and events without replacement are independent.

Rationale for Option D: **Key** – The student recognizes that the probability of picking the second crayon is affected by the first crayon because it was not replaced, and therefore the probability of picking a certain color crayon is increased, so it is dependent.

Sample Response: 1 point



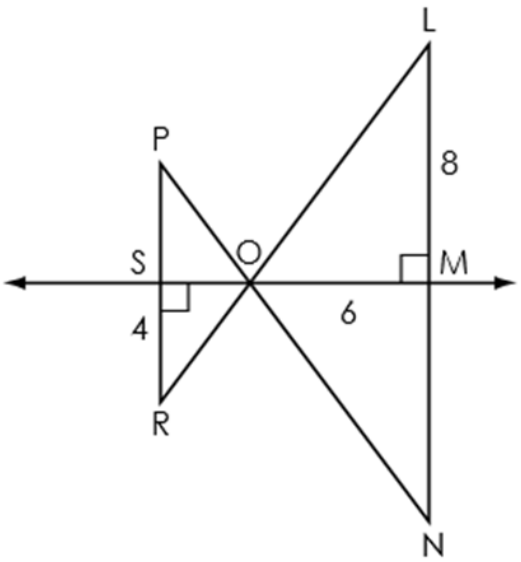
Geometry
Spring 2025 Item Release

Question 5

Question and Scoring Guidelines

Question 5

Triangles RPO and NLO are shown where $\overleftrightarrow{SM}$ bisects $\overline{PR}$ and $\overline{LN}$.



What is the area, in square units, of $\triangle RPO$?

square units



Points Possible: 1

Content Cluster: Prove and apply theorems both formally and informally involving similarity using a variety of methods.

Content Standard: Use congruence and similarity criteria for triangles to solve problems and to justify relationships in geometric figures that can be decomposed into triangles. *(G.SRT.5)*

Depth of Knowledge: Level 3

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student enters 12, providing evidence of the ability to use congruence and similarity criteria for triangles to solve problems in geometric figures that can be decomposed into triangles.

Exemplar Response

- 12 square units
  or
- any equivalent value

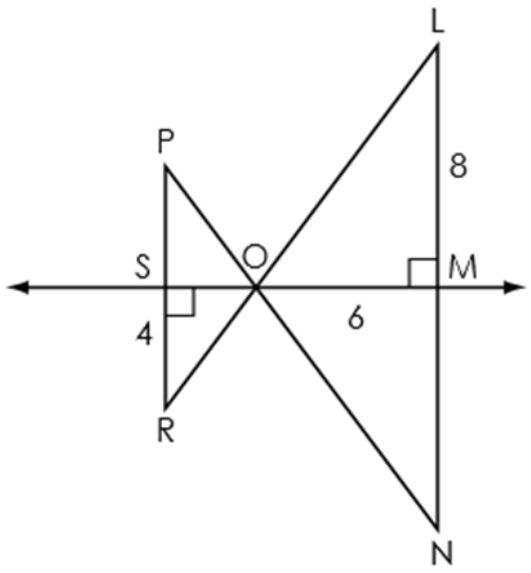
Geometry
Spring 2025 Item Release

Question 5

Sample Responses

Sample Response: 1 point

Triangles RPO and NLO are shown where $\overleftrightarrow{SM}$ bisects $\overline{PR}$ and $\overline{LN}$.



What is the area, in square units, of $\triangle RPO$?

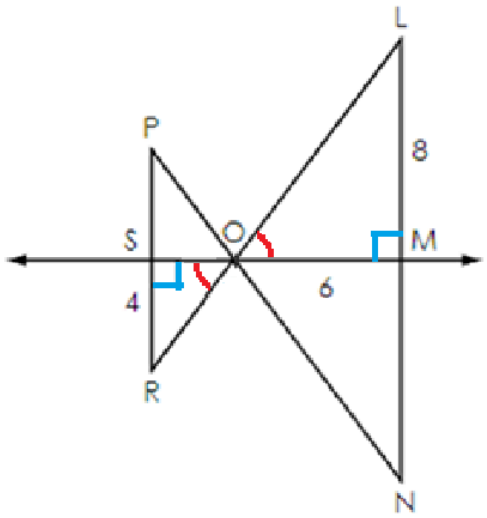
12 *square units*



Notes on Scoring

This response receives full credit (1 point) because it correctly provides the area of triangle RPO.

The student uses the Angle-Angle Similarity Theorem to show △OSR and △OML are similar, identifying that ∠SOR ≅ ∠MOL because they are vertical angles, and ∠OSR ≅ ∠OML because they are right angles.



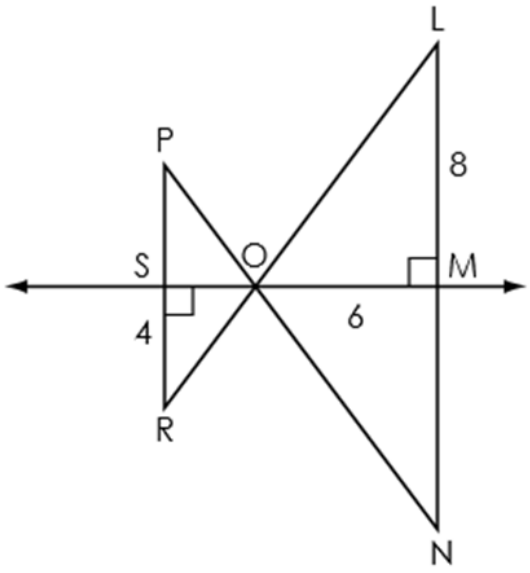
Since the triangles are similar, $\frac{OS}{RS} = \frac{OM}{LM}$, so $\frac{OS}{4} = \frac{6}{8}$, or $OS = 3$.

The student understands that $\overline{PS}$ is the same length as $\overline{RS}$ because $\overleftrightarrow{SM}$ bisects $\overline{PR}$. Since PS and RS are both equal to 4, PR = 8.

The student then calculates the area of △RPO using the formula $A = \frac{1}{2}bh$, where b is the base, or 8, and h is the height, or 3. The area of △RPO is $\frac{1}{2}(8)(3)$, or 12.

Sample Response: 0 points

Triangles RPO and NLO are shown where $\overleftrightarrow{SM}$ bisects $\overline{PR}$ and $\overline{LN}$.



What is the area, in square units, of $\triangle RPO$?

6 *square units*

1	2	3
4	5	6
7	8	9
	0	
.	-	

Notes on Scoring

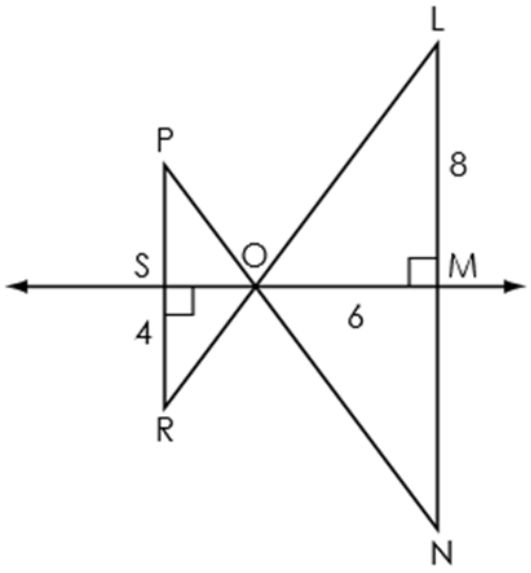
This response receives no credit (0 points) because it does not provide the correct area of triangle RPO.

The student may use the Angle-Angle Similarity Theorem to show △OSR and △OML are similar, identifying that ∠SOR ≅ ∠MOL because they are vertical angles, and ∠OSR ≅ ∠OML because they are right angles.

Since the triangles are similar $\frac{\mathrm{OS}}{\mathrm{RS}} = \frac{\mathrm{OM}}{\mathrm{LM}}$ so $\frac{\mathrm{OS}}{4} = \frac{6}{8}$, or $\mathrm{OS} = 3$. The student may then calculate the area of △SPO, instead of △RPO, using the formula $A = \frac{1}{2}bh$, where b is the base, or 4, and h is the height, or 3, resulting in $\frac{1}{2}(4)(3)$, or 6.

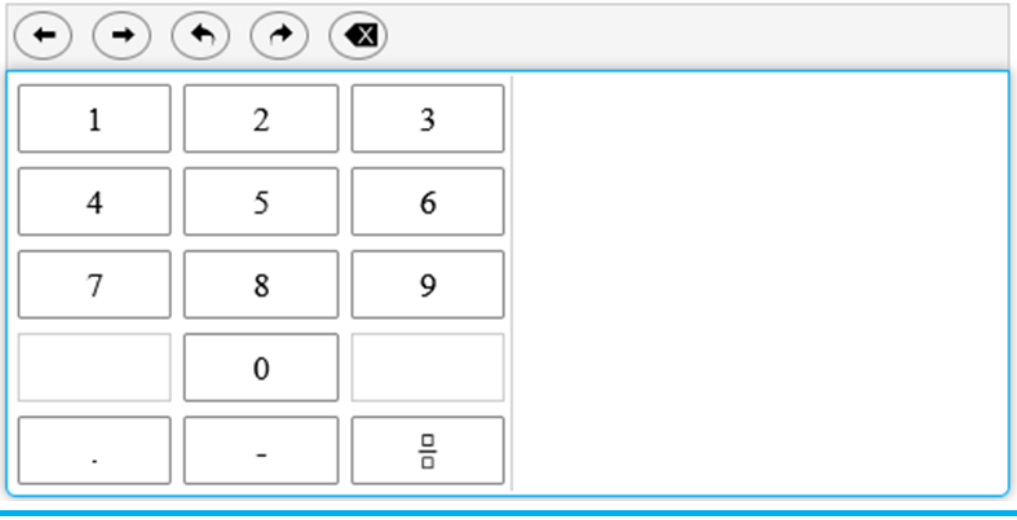
Sample Response: 0 points

Triangles RPO and NLO are shown where $\overleftrightarrow{SM}$ bisects $\overline{PR}$ and $\overline{LN}$.



What is the area, in square units, of $\triangle RPO$?

8 *square units*



Notes on Scoring

This response receives no credit (0 points) because it does not provide the correct area of triangle RPO.

The student may understand that △OSR and △OML are similar but may think that since $\mathrm{OM} = \mathrm{ML} - 2$, $\mathrm{OS} = \mathrm{SR} - 2$. So, the student calculates $4 - 2 = 2$ for OS.

The student understands that $\overline{\mathrm{PS}}$ is the same length as $\overline{\mathrm{RS}}$ because $\overleftrightarrow{\mathrm{SM}}$ bisects $\overline{\mathrm{PR}}$. Since PS and RS are both equal to 4, PR = 8.

The student then calculates the area of △RPO using the formula $A = \frac{1}{2}bh$, where b is the base, or 8, and h is the incorrect height, 2. The area of △RPO is $\frac{1}{2}(8)(2)$, or 8.

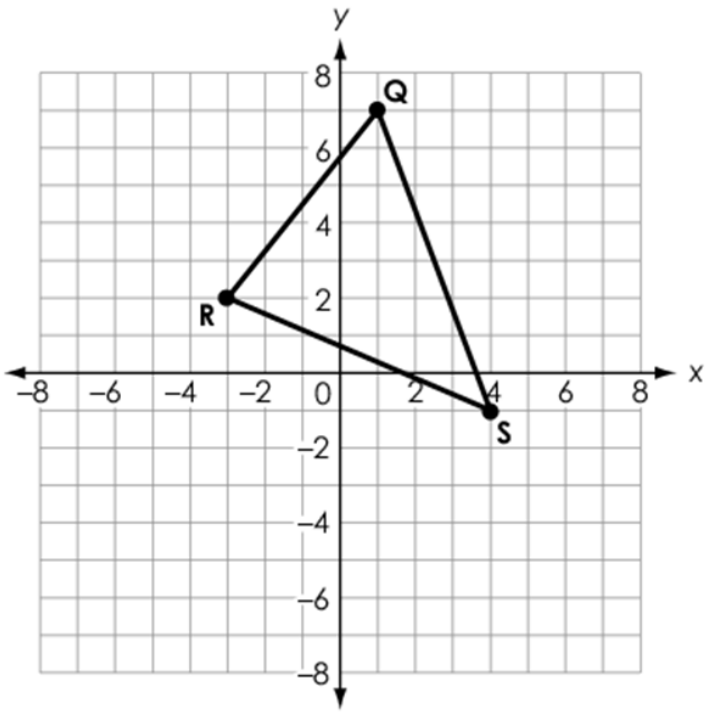
Geometry
Spring 2025 Item Release

Question 11

Question and Scoring Guidelines

Question 11

Triangle QRS is transformed using the rule $(-x, y + 2)$ to create triangle Q′R′S′



What are the *x*- and *y*-coordinates of the vertices of triangle Q′R′S′?

Q' (,)

R' (,)

S' (,)

Points Possible: 1

Content Cluster: Experiment with transformations in the plane.

Content Standard: Represent transformations in the plane using, e.g., transparencies and geometry software; describe transformations as functions that take points in the plane as inputs and give other points as outputs. Compare transformations that preserve distance and angle to those that do not (e.g., translation versus horizontal stretch). *(G.CO.2)*

Depth of Knowledge: Level 1

Modeling and Reasoning: No

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student identifies the correct coordinates, providing evidence of the ability to use a function to transform coordinates in a figure.

Exemplar Response

- Q' (–1, 9)
  R' (3, 4)
  S' (–4, 1)
  or
- any equivalent values

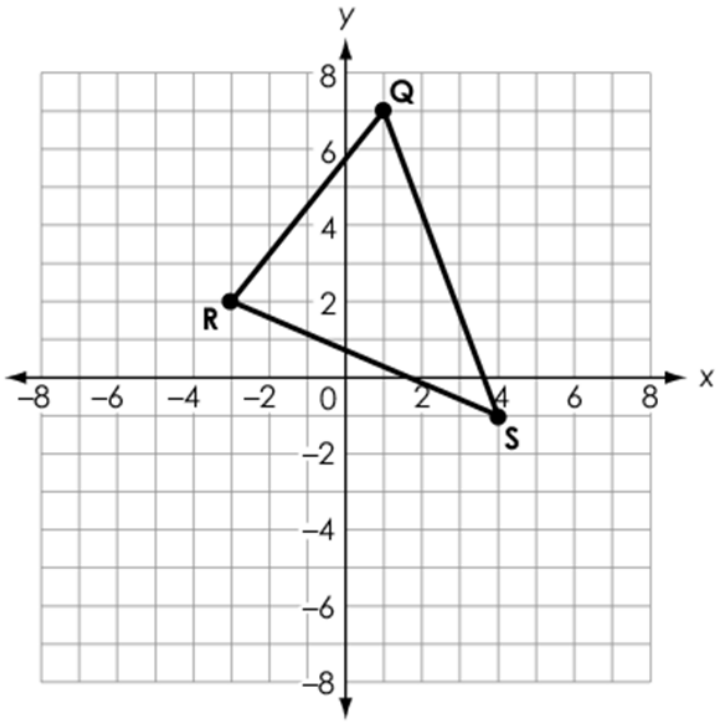
Geometry
Spring 2025 Item Release

Question 11

Sample Responses

Sample Response: 1 point

Triangle QRS is transformed using the rule $(-x, y + 2)$ to create triangle Q′R′S′.



What are the *x*- and *y*-coordinates of the vertices of triangle Q′R′S′?

Q' (−1 , 9)

R' (3 , 4)

S' (−4 , 1)

Notes on Scoring

This response receives full credit (1 point) because it shows the correct coordinates of the transformed triangle Q′R′S′.

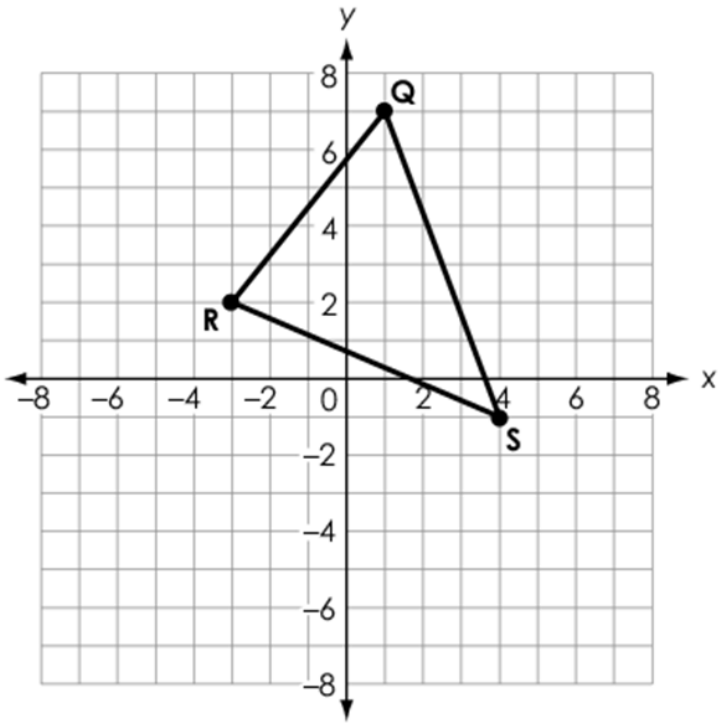
The student understands the rule will be used on the coordinates for the given triangle.

They take the opposite of the x-coordinate and add 2 to the *y*-coordinate.

Original Coordinate		Rule		Calculation		New Coordinate
Q: $(1, 7)$	$\longrightarrow$	$(-x,\ y+2)$	$\longrightarrow$	$(-1,\ 7+2)$	$\longrightarrow$	Q′: $(-1, 9)$
R: $(-3, 2)$	$\longrightarrow$	$(-x,\ y+2)$	$\longrightarrow$	$(-(-3),\ 2+2)$	$\longrightarrow$	R′: $(3, 4)$
S: $(4, -1)$	$\longrightarrow$	$(-x,\ y+2)$	$\longrightarrow$	$(-4,\ -1+2)$	$\longrightarrow$	S′: $(-4, 1)$

Sample Response: 0 points

Triangle QRS is transformed using the rule $(-x, y + 2)$ to create triangle Q′R′S′



What are the *x*- and *y*-coordinates of the vertices of triangle Q′R′S′?

Q′ (−1 , 5)

R′ (3 , 0)

S′ (−4 , −3)

Notes on Scoring

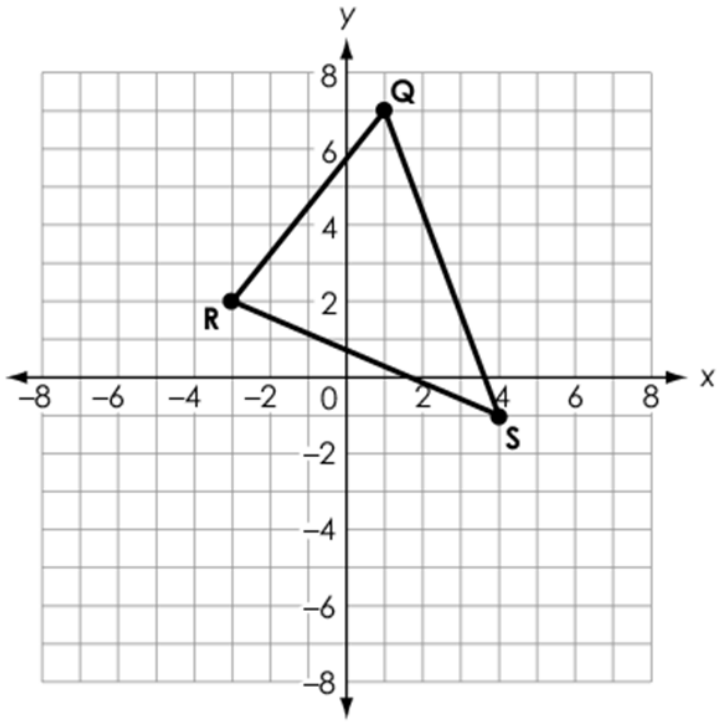
This response receives no credit (0 points) because it does not show the correct coordinates of the transformed triangle Q'R'S'.

The student may think the rule was used on the coordinates for the new triangle to create the original triangle. The student applies the inverse of the rule to the original coordinates to determine the new coordinates, taking the opposite of the x-coordinate and subtracting 2 from the *y*-coordinate.

Original Coordinate	"Inverse" Rule		Calculation		New Coordinate
Q: $(1, 7)$	$(-x,\ y-2)$	$\longrightarrow$	$(-1,\ 7-2)$	$\longrightarrow$	Q′: $(-1, 5)$
R: $(-3, 2)$	$(-x,\ y-2)$	$\longrightarrow$	$(-(-3),\ 2-2)$	$\longrightarrow$	R′: $(3, 0)$
S: $(4, -1)$	$(-x,\ y-2)$	$\longrightarrow$	$(-4,\ -1-2)$	$\longrightarrow$	S′: $(-4, -3)$

Sample Response: 0 points

Triangle QRS is transformed using the rule $(-x, y + 2)$ to create triangle Q′R′S′



What are the *x*- and *y*-coordinates of the vertices of triangle Q′R′S′?

Q′ (−1 , 9)

R′ (−3 , 4)

S′ (−4 , −1)

Notes on Scoring

This response receives no credit (0 points) because it does not show the correct coordinates of the transformed triangle Q'R'S'.

The student may understand the rule will be used on the coordinates for the given triangle but may think the –x in the rule means the x-coordinate is always going to be negative.

Original Coordinate		Rule		Calculation		New Coordinate
Q: $(1, 7)$	$\longrightarrow$	$(-x,\ y+2)$	$\longrightarrow$	$(-1,\ 7+2)$	$\longrightarrow$	Q': $(-1, 9)$
R: $(-3, 2)$	$\longrightarrow$	$(-x,\ y+2)$	$\longrightarrow$	$(-3,\ 2+2)$	$\longrightarrow$	R': $(-3, 4)$
S: $(4, -1)$	$\longrightarrow$	$(-x,\ y+2)$	$\longrightarrow$	$(-4,\ -1+2)$	$\longrightarrow$	S': $(-4, 1)$

Geometry
Spring 2025 Item Release

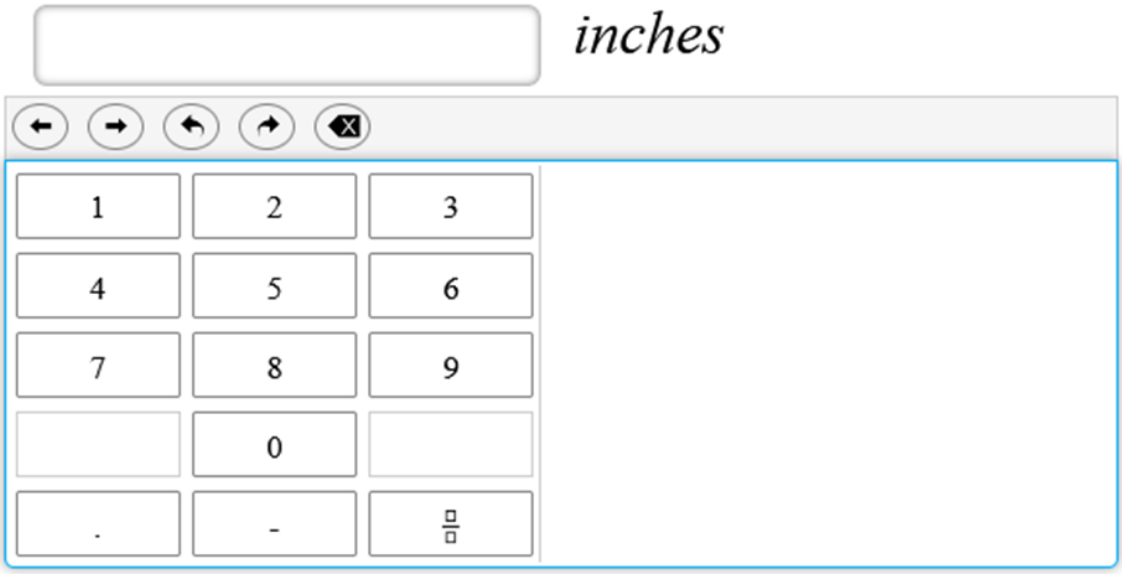
Question 12

Question and Scoring Guidelines

Question 12

A company makes cylindrical containers with a height of 10 inches. The volume of each container is less than 600 cubic inches and the surface area is greater than 200 square inches.

What is a possible radius, in inches, of a cylindrical container that the company makes?



Points Possible: 1

Content Cluster: Apply geometric concepts in modeling situations.

Content Standard: Apply geometric methods to solve design problems (e.g., designing an object or structure to satisfy physical constraints or minimize cost; working with typographic grid systems based on ratios). *(G.MG.3)*

Depth of Knowledge: Level 2

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student enters 3, providing evidence of the ability to apply geometric methods to solve design problems.

Exemplar Response

- 3 inches
  or
- any value greater than 2.53 and less than 4.3713019

Geometry
Spring 2025 Item Release

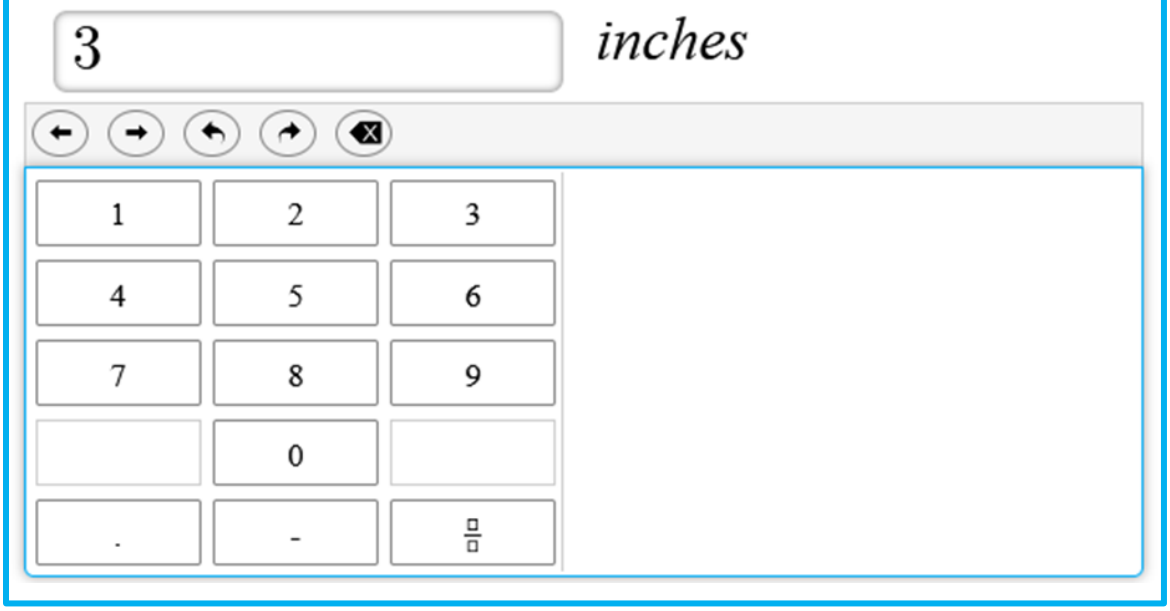
Question 12

Sample Responses

Sample Response: 1 point

A company makes cylindrical containers with a height of 10 inches. The volume of each container is less than 600 cubic inches and the surface area is greater than 200 square inches.

What is a possible radius, in inches, of a cylindrical container that the company makes?



Notes on Scoring

This response receives full credit (1 point) because it identifies a possible radius of the cylindrical container.

The student substitutes the given values into the formula for the volume of a cylinder.

Maximum Volume of 600

$V = \pi r^2 h$

$600 \geq \pi r^2(10)$

$\frac{60}{\pi} \geq r^2$

$\sqrt{\frac{60}{\pi}} \geq r$

$4.37 \geq r$

The student identifies a value less than 4.37, 3, for the radius, so that the volume will be less than 600 cubic inches.

The student substitutes $r = 3$ and $h = 10$ into the formula for the surface area of a cylinder, $SA = 2\pi rh + 2\pi r^2$, to check that the surface area is greater than 200 square inches.

Minimum Surface Area of 200

$SA = 2\pi rh + 2\pi r^2$

$200 \leq 2\pi(3)10 + 2\pi(3)^2$

$200 \leq 2\pi(30) + 2\pi(9)$

$200 \leq \quad 60\pi + 18\pi$

$200 \leq 188.5 + 66.55$

$200 \leq 255.05$

Since the volume is less than 600 cubic inches and the surface area is greater than 200 square inches, 3 is a correct value for the radius.

Sample Response: 1 point

A company makes cylindrical containers with a height of 10 inches. The volume of each container is less than 600 cubic inches and the surface area is greater than 200 square inches.

What is a possible radius, in inches, of a cylindrical container that the company makes?

4 *inches*



Notes on Scoring

This response receives full credit (1 point) because it identifies a possible radius of the cylindrical container.

The student substitutes the given values into the formula provided on the reference sheet for the volume of a cylinder.

Maximum Volume of 600

$V = \pi r^2 h$

$600 \geq \pi r^2(10)$

$\frac{60}{\pi} \geq r^2$

$\sqrt{\frac{60}{\pi}} \geq r$

$4.37 \geq r$

The student identifies a value less than 4.37, 4, for the radius, so that the volume will be less than 600 cubic inches. The student chooses a value close to the maximum for the radius because the surface area needs to be greater than 200 square inches, and the larger the radius, the larger the surface area.

The student substitutes $r = 4$ and $h = 10$ into the formula for the surface area of a cylinder, $SA = 2\pi rh + 2\pi r^2$, to check that the surface area is greater than 200 square inches.

Minimum Surface Area of 200

$SA = 2\pi rh + 2\pi r^2$

$200 \leq 2\pi(4)10 + 2\pi(4)^2$

$200 \leq 2\pi(40) + 2\pi(16)$

$200 \leq 80\pi + 32\pi$

$200 \leq 251.33 + 100.53$

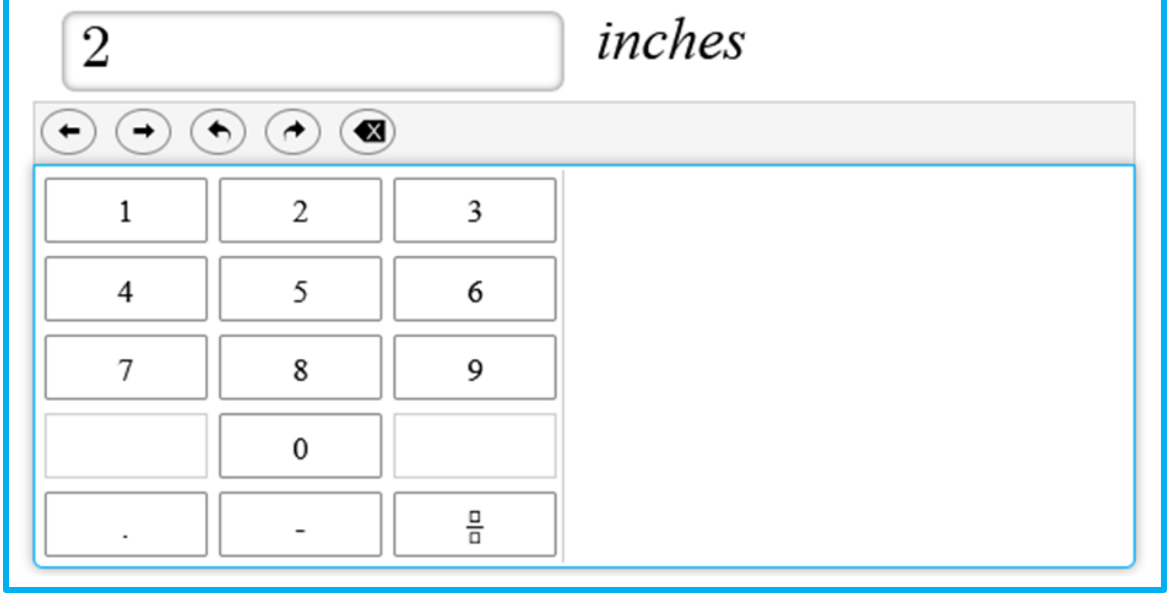
$200 \leq 351.86$

Since the volume is less than 600 cubic inches and the surface area is greater than 200 square inches, 4 is a correct value for the radius.

Sample Response: 0 points

A company makes cylindrical containers with a height of 10 inches. The volume of each container is less than 600 cubic inches and the surface area is greater than 200 square inches.

What is a possible radius, in inches, of a cylindrical container that the company makes?



Notes on Scoring

This response receives no credit (0 points) because it does not identify a possible radius of the cylindrical container.

The student may substitute the given values into the formula for the volume of a cylinder to determine a possible value for the radius.

Maximum Volume of 600

$V = \pi r^2 h$

$600 \geq \pi r^2 (10)$

$\frac{60}{\pi} \geq r^2$

$\sqrt{\frac{60}{\pi}} \geq r$

$4.37 \geq r$

The student identifies a value less than 4.37, 2, for the radius, so that the volume will be less than 600 cubic inches.

The student, however, may not check that the surface area is greater than 200 square inches.

Minimum Surface Area of 200

$SA = 2\pi rh + 2\pi r^2$

$200 \leq 2\pi(2)10 + 2\pi(2)^2$

$200 \leq 2\pi(20) + 2\pi(4)$

$200 \leq \quad 40\pi + 8\pi$

$200 \leq 125.66 + 25.13$

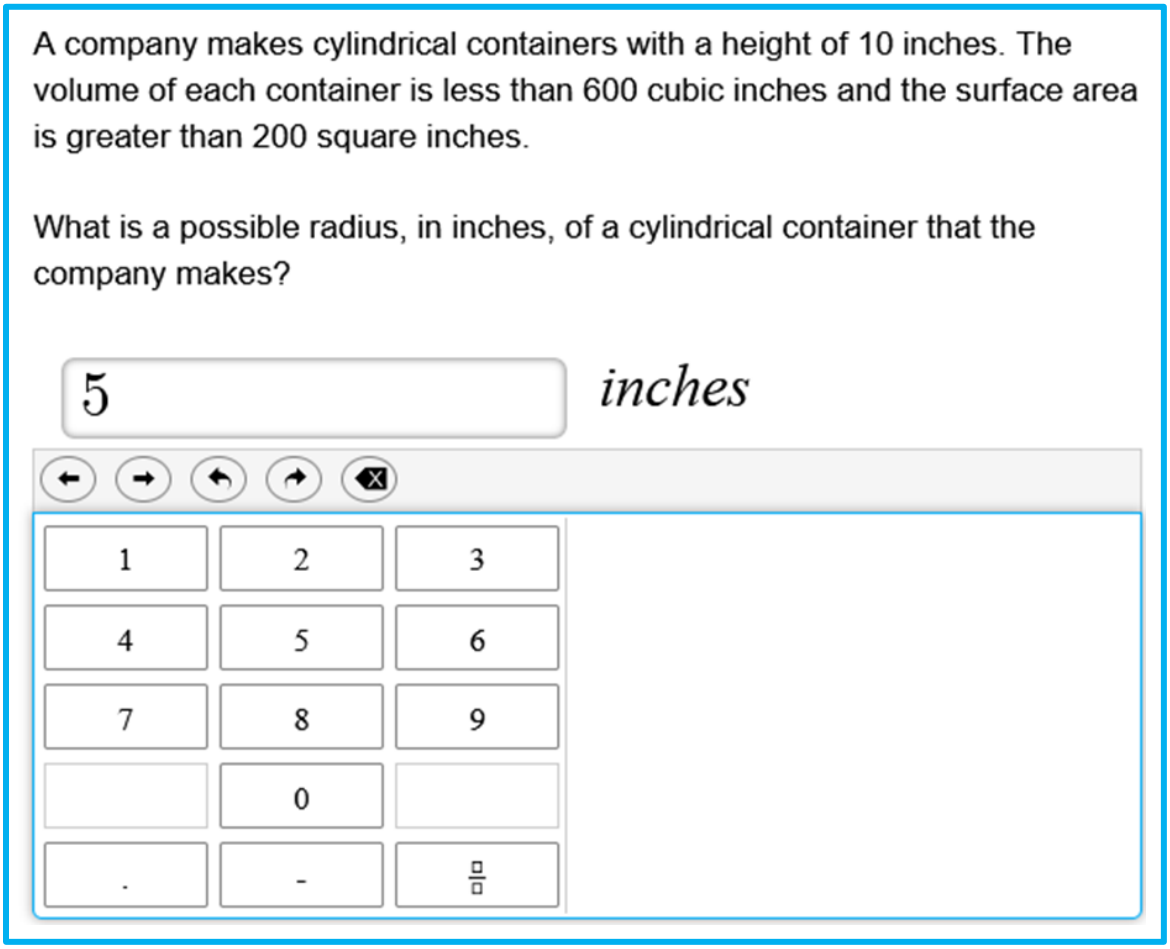
$200 \leq 150.79$

Using a radius of 2 gives a surface area of 150.79, which is less than 200 square inches, so 2 is not a correct value for the radius.

Sample Response: 0 points

A company makes cylindrical containers with a height of 10 inches. The volume of each container is less than 600 cubic inches and the surface area is greater than 200 square inches.

What is a possible radius, in inches, of a cylindrical container that the company makes?



Notes on Scoring

This response receives no credit (0 points) because it does not identify a possible radius of the cylindrical container.

The student may substitute the given values into the formula for the volume of a cylinder, but the student may think that 600 should be less than or equal to the volume.

Maximum Volume of 600

$V = \pi r^2 h$

$600 \leq \pi r^2 (10)$

$\frac{60}{\pi} \leq r^2$

$\sqrt{\frac{60}{\pi}} \leq r$

$4.37 \leq r$

The student identifies a value **greater than** 4.37, 5, for the radius, thinking that the volume will be less than 600 cubic inches.

The student substitutes $r = 5$ and $h = 10$ into the formula for the surface area of a cylinder, $SA = 2\pi rh + 2\pi r^2$, to check that the surface area is greater than 200 square inches.

Minimum Surface Are of 200

$SA = 2\pi rh + 2\pi r^2$

$200 \leq 2\pi(5)10 + 2\pi(5)^2$

$200 \leq 2\pi(50) + 2\pi(25)$

$200 \leq 100\pi + 50\pi$

$200 \leq 150\,\pi$

$200 \leq 471.24$

While the surface area is greater than 200 square units, the volume is not less than 600 cubic units, so $r = 5$ is not a correct value for the radius.

Geometry
Spring 2025 Item Release

Question 20

Question and Scoring Guidelines

Question 20

Two figures are shown, with side lengths in units.

Figure 1

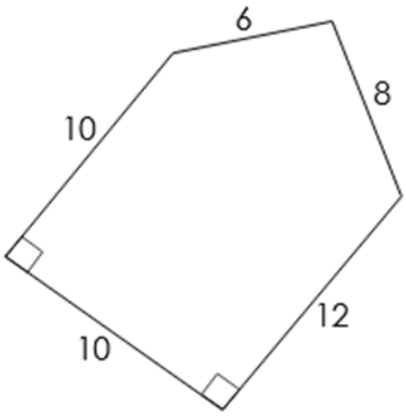
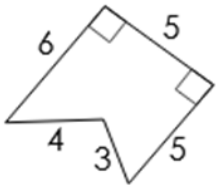


Figure 2



What is true about the two figures?

(A) Figure 1 and Figure 2 are similar by AA criterion.

(B) Figure 1 and Figure 2 are similar because the sides of Figure 2 are proportional to the sides of Figure 1.

(C) Figure 1 and Figure 2 are not similar because the orientation of Figure 2 is different from the orientation of Figure 1.

(D) Figure 1 and Figure 2 are not similar because Figure 1 cannot be mapped onto Figure 2 using similarity transformations.

Points Possible: 1

Content Cluster: Understand similarity in terms of similarity transformations.

Content Standard: Given two figures, use the definition of similarity in terms of similarity transformations to decide if they are similar; explain using similarity transformations the meaning of similarity for triangles as the equality of all corresponding pairs of angles and the proportionality of all corresponding pairs of sides. *(G.SRT.2)*

Depth of Knowledge: Level 2

Modeling and Reasoning: No

Scoring Guidelines

Rationale for Option A: This is incorrect. The student may not realize that the AA criterion only holds for triangles, not polygons. Similar polygons are those that can be mapped onto each other using similarity transformations.

Rationale for Option B: This is incorrect. The student may not realize that sides being proportional is not a sufficient criterion for similarity with polygons of more than 3 sides, or that while the sides in these figures are proportional, they are arranged in different ways so that the figures do not have similar shapes.

Rationale for Option C: This is incorrect. The student may recognize that the figures are not similar but may think it is solely because of the difference in orientation, not realizing that similar shapes can be oriented differently since rotation is a similarity transformation.

Rationale for Option D: **Key** – The student recognizes that for two figures to be similar one figure must be able to be mapped onto the other and recognizes that Figure 1 and Figure 2 are not the same shape because Figure 1 is a convex shape and Figure 2 is a concave shape.

Sample Response: 1 point

Two figures are shown, with side lengths in units.

Figure 1

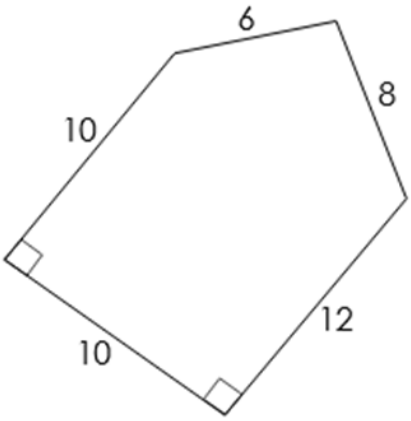
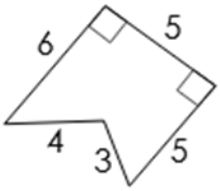


Figure 2



What is true about the two figures?

- (A) Figure 1 and Figure 2 are similar by AA criterion.
- (B) Figure 1 and Figure 2 are similar because the sides of Figure 2 are proportional to the sides of Figure 1.
- (C) Figure 1 and Figure 2 are not similar because the orientation of Figure 2 is different from the orientation of Figure 1.
- ● Figure 1 and Figure 2 are not similar because Figure 1 cannot be mapped onto Figure 2 using similarity transformations.

Geometry
Spring 2025 Item Release

Question 21

Question and Scoring Guidelines

Question 21

The probabilities of two independent events are given.

$P(A) = 0.82$
$P(B) = 0.60$

What is $P(A \cap B)$?



Points Possible: 1

Content Cluster: Understand independence and conditional probability, and use them to interpret data.

Content Standard: Understand that two events A and B are independent if and only if the probability of A and B occurring together is the product of their probabilities, and use this characterization to determine if they are independent. *(S.CP.2)*

Depth of Knowledge: Level 1

Modeling and Reasoning: No

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student calculates the probability as 0.492, providing evidence of understanding the probability of independent events is equal to the product of their individual probabilities.

Exemplar Response

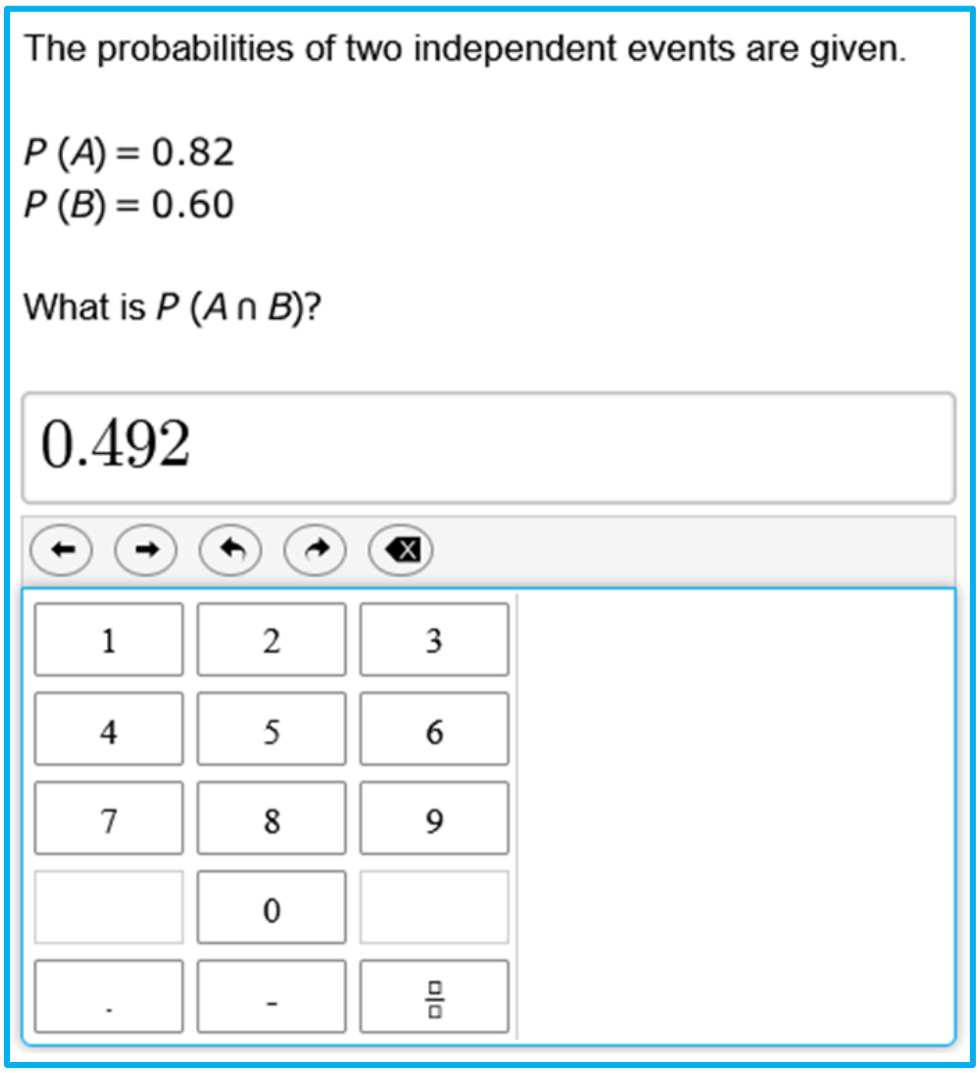
- 0.492
  or
- any equivalent value

Geometry
Spring 2025 Item Release

Question 21

Sample Responses

Sample Response: 1 point

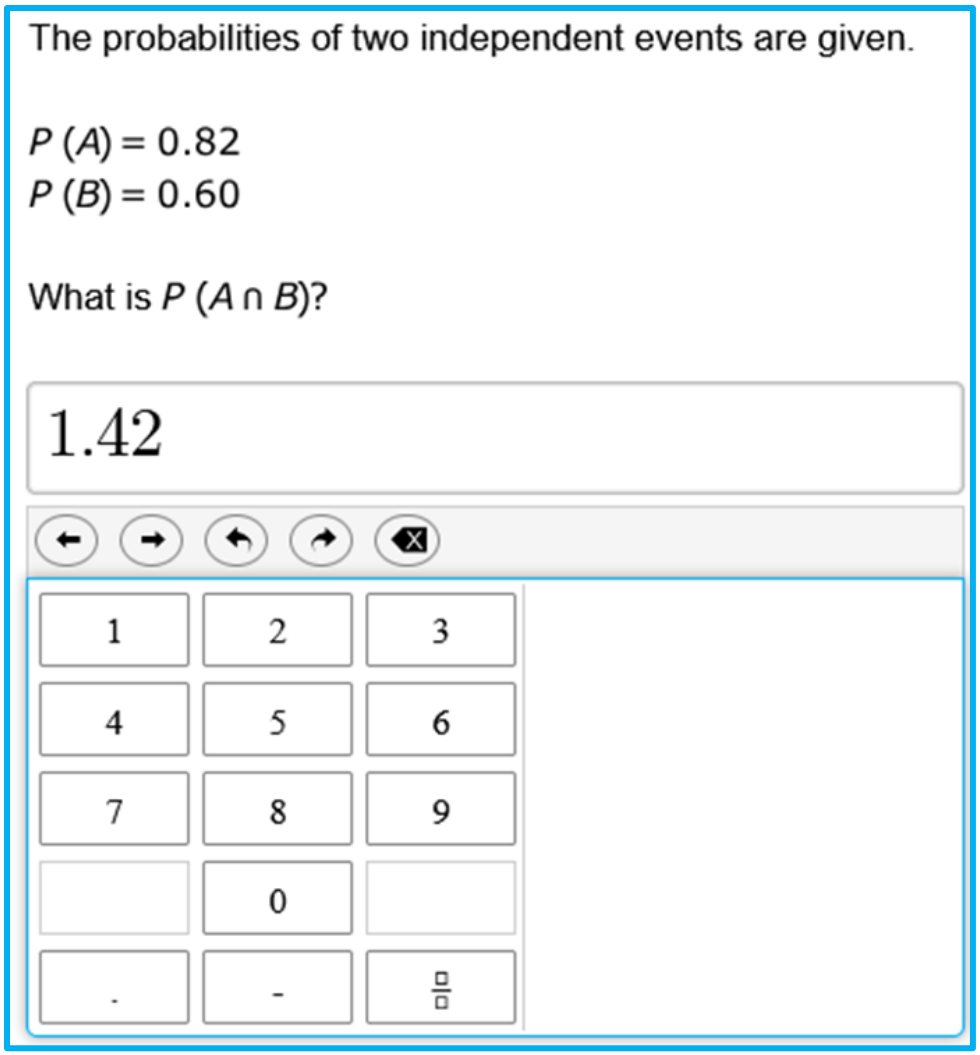


Notes on Scoring

This response receives full credit (1 point) because it provides the correct value of $P(A \cap B)$.

The student understands that since $P(A)$ and $P(B)$ are independent, $P(A \cap B) = P(A) \times P(B)$, and calculates $0.82 \times 0.60 = 0.492$.

Sample Response: 0 points

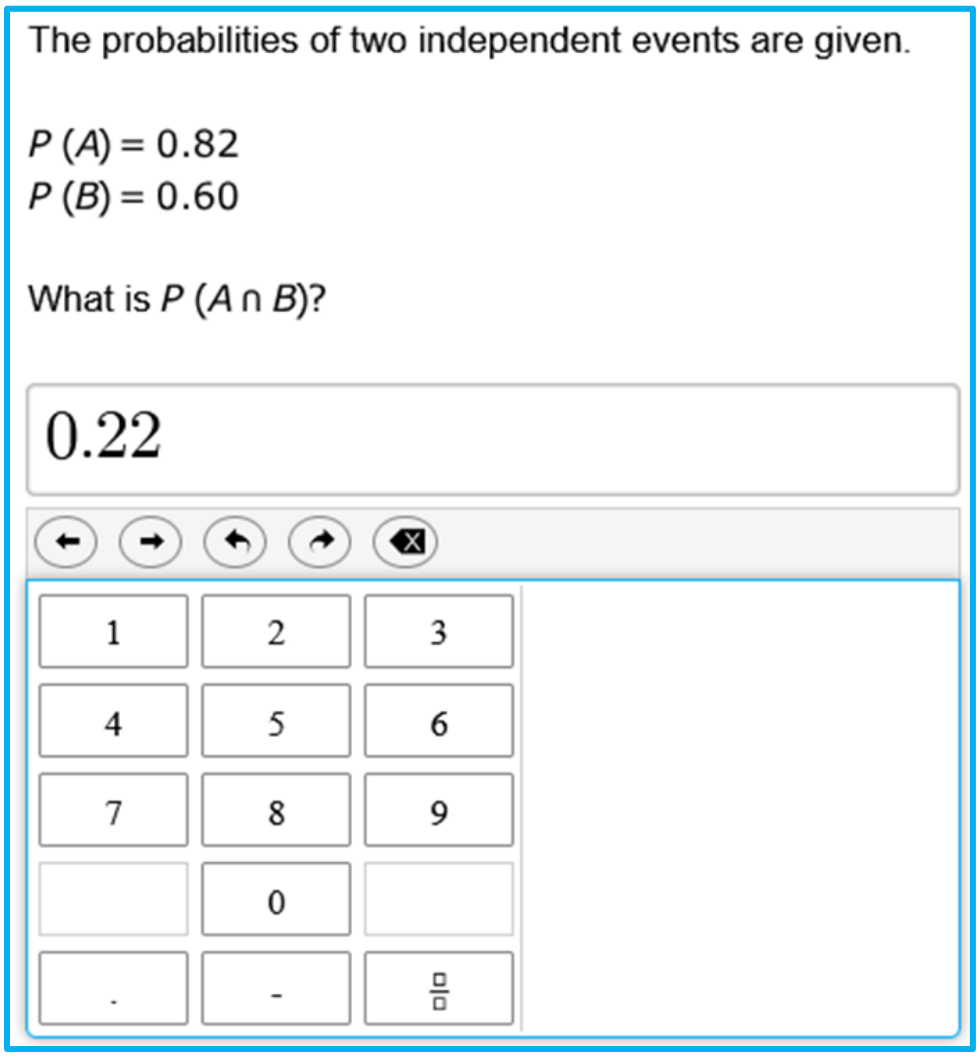


Notes on Scoring

This response receives no credit (0 points) because it does not provide the correct value of $P(A \cap B)$.

The student may think that since the ∩ symbol means "and" that $P(A \cap B) = P(A) + P(B)$ and calculates 0.82 + 0.60 = 1.42.

Sample Response: 0 points



Notes on Scoring

This response receives no credit (0 points) because it does not provide the correct value of $P(A \cap B)$.

The student may think that $P(A \cap B) = P(A) - P(B)$ and calculates $0.82 - 0.60 = 0.22$.

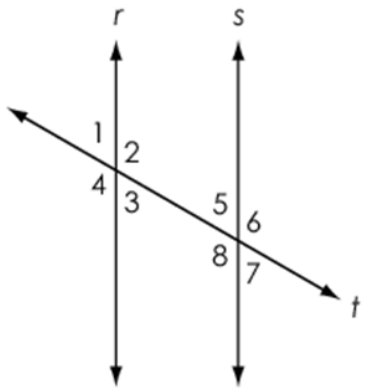
Geometry
Spring 2025 Item Release

Question 22

Question and Scoring Guidelines

Question 22

Parallel lines r and s are cut by a transversal t as shown.



Jasmine wants to prove that corresponding angles are congruent. She writes the partial paragraph proof shown.

Since vertical angles are congruent, $\angle 2 \cong \angle 4$. Also, since alternate exterior angles are congruent when parallel lines are cut by a transversal, $\angle 4 \cong \angle 6$.

Which statement could complete Jasmine's proof?

Ⓐ Then, $\angle 2 \cong \angle 6$ by the transitive property.

Ⓑ Then, $\angle 4 \cong \angle 8$ by the transitive property.

Ⓒ Then, $\angle 2 \cong \angle 8$ since alternate interior angles are congruent.

Ⓓ Then, $\angle 6 \cong \angle 8$ since alternate interior angles are congruent.

Points Possible: 1

Content Cluster: Prove geometric theorems both formally and informally using a variety of methods.

Content Standard: Prove and apply theorems about lines and angles. Theorems include but are not restricted to the following: vertical angles are congruent; when a transversal crosses parallel lines, alternate interior angles are congruent and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints. *(G.CO.9)*

Depth of Knowledge: Level 2

Modeling and Reasoning: Yes

Scoring Guidelines

Rationale for Option A: **Key** – The student identifies the correct conclusion. Since angle 4 is congruent to angle 6 and to angle 2, the transitive property states that angle 2 is congruent to angle 6. Since angles 2 and 6 are corresponding angles, this completes the proof.

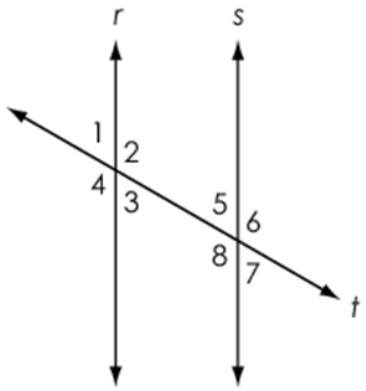
Rationale for Option B: This is incorrect. The student may realize that angles 4 and 8 are corresponding angles but misses that since no information about angle 8 is given in the partial proof, it cannot be concluded that angle 4 is congruent to angle 8.

Rationale for Option C: This is incorrect. The student may realize that this fact could be used to extend the proof and show that corresponding angles 4 and 8 are congruent but misses that this statement cannot complete a proof that corresponding angles are congruent. Another statement would be necessary to complete that proof.

Rationale for Option D: This is incorrect. The student may think that since vertical angles 2 and 4 were mentioned in the partial proof, and angles 6 and 8 are also vertical angles, a statement about their congruence should be included in the proof.

Sample Response: 1 point

Parallel lines r and s are cut by a transversal t as shown.



Jasmine wants to prove that corresponding angles are congruent. She writes the partial paragraph proof shown.

Since vertical angles are congruent, $\angle 2 \cong \angle 4$. Also, since alternate exterior angles are congruent when parallel lines are cut by a transversal, $\angle 4 \cong \angle 6$.

Which statement could complete Jasmine's proof?

- ● Then, $\angle 2 \cong \angle 6$ by the transitive property.
- Ⓑ Then, $\angle 4 \cong \angle 8$ by the transitive property.
- Ⓒ Then, $\angle 2 \cong \angle 8$ since alternate interior angles are congruent.
- Ⓓ Then, $\angle 6 \cong \angle 8$ since alternate interior angles are congruent.

Geometry
Spring 2025 Item Release

Question 24

Question and Scoring Guidelines

Question 24

Which statement about triangles must be true?

Ⓐ A right triangle is always a scalene triangle.

Ⓑ A right triangle is always an isosceles triangle.

Ⓒ An isosceles triangle is always an acute triangle.

Ⓓ An equilateral triangle is always an acute triangle.

Points Possible: 1

Content Cluster: Classify and analyze geometric figures.

Content Standard: Classify two-dimensional figures in a hierarchy based on properties. *(G.CO.14)*

Depth of Knowledge: Level 2

Modeling and Reasoning: No

Scoring Guidelines

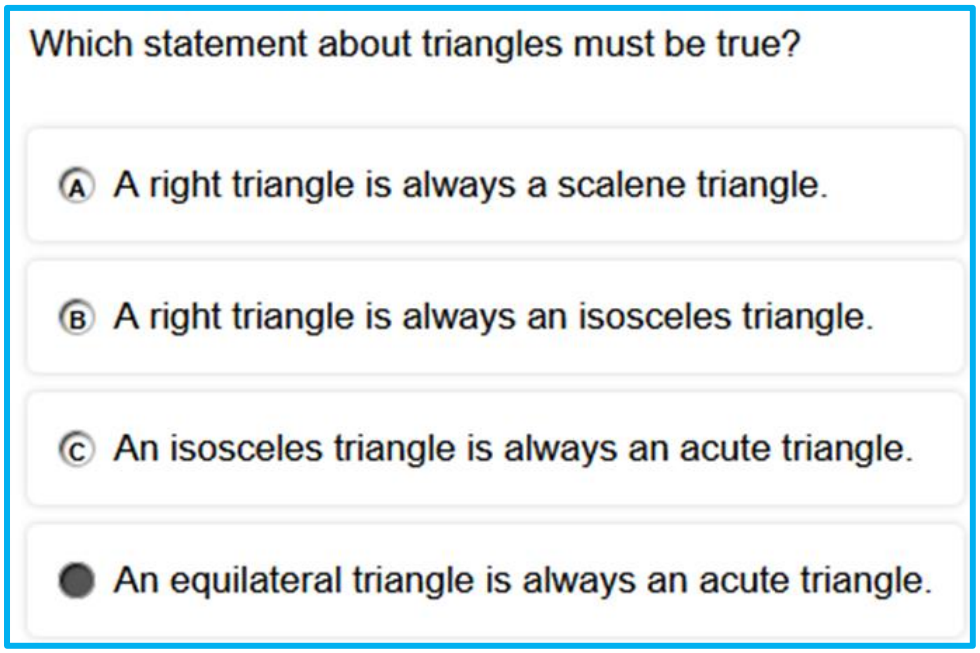
Rationale for Option A: This is incorrect. The student may disregard the case when a right triangle has two legs of equal lengths, which would make it isosceles rather than scalene.

Rationale for Option B: This is incorrect. The student may disregard the case when a right triangle has two legs of different lengths, which would make it scalene, rather than isosceles.

Rationale for Option C: This is incorrect. The student may disregard the case when an isosceles triangle has an apex angle greater than 90 degrees, which would make the triangle obtuse rather than acute.

Rationale for Option D: **Key** – The student correctly identifies that since all of the angles of an equilateral triangle measure 60 degrees, all of the angles are acute, making the triangle acute.

Sample Response: 1 point



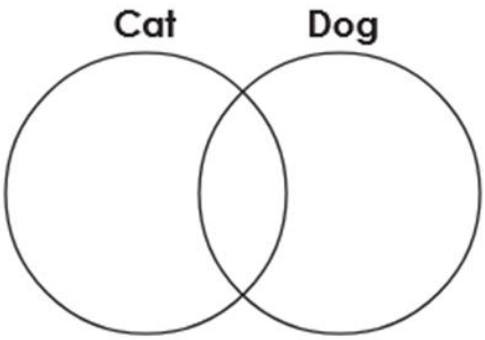
Geometry
Spring 2025 Item Release

Question 25

Question and Scoring Guidelines

Question 25

Rodas asks pet owners whether they own a cat or a dog and records the data in the Venn diagram shown.



Let C be the event that a pet owner owns a cat and D be the event that a pet owner owns a dog.

Which expression can be used to find the probability that a pet owner owns a cat or a dog?

(A) $P(C) + P(D) - P(C \cup D)$

(B) $P(C) + P(D) - P(C \cap D)$

(C) $P(C) + P(D) + P(C \cup D)$

(D) $P(C) + P(D) + P(C \cap D)$

Points Possible: 1

Content Cluster: Use the rules of probability to compute probabilities of compound events in a uniform probability model.

Content Standard: Apply the Addition Rule,
$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$, and interpret the answer in terms of the model. *(S.CP.7)*

Depth of Knowledge: Level 1

Modeling and Reasoning: Yes

Scoring Guidelines

Rationale for Option A: This is incorrect. The student may confuse the ∪ and ∩ symbols and their interpretation (union and intersection).

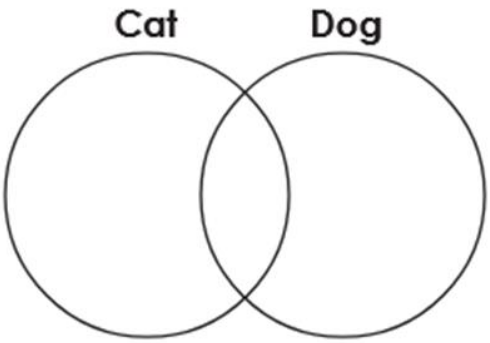
Rationale for Option B: **Key** – The student recognizes that adding all the cat owners plus all of the dog owners counts all the people who own a cat or a dog, but this also counts the pet owners who own a cat *and* a dog twice. The student also recognizes that the pet owners who own both a cat and a dog are represented by the intersection of the two circles. The student understands to subtract the pet owners who own both a cat and a dog from the sum of the pet owners who own a cat and the pet owners who own a dog to only count those who own a cat and a dog one time.

Rationale for Option C: This is incorrect. The student may know that the Addition rule can be used to find the probability of a pet owner owning a cat or a dog but thinks the formula for the Addition rule adds the union of *C* and *D* to the sum of the individual probabilities of *C* and *D* instead of subtracting the intersection of *C* and *D* from the sum of the individual probabilities.

Rationale for Option D: This is incorrect. The student may know that the Addition rule can be used to find the probability of a pet owner owning a cat or a dog but thinks the formula for the Addition rule adds the intersection of *C* and *D* to the sum of the individual probabilities of *C* and *D* instead of subtracting the intersection of *C* and *D* from the sum of the individual probabilities.

Sample Response: 1 point

Rodas asks pet owners whether they own a cat or a dog and records the data in the Venn diagram shown.



Let C be the event that a pet owner owns a cat and D be the event that a pet owner owns a dog.

Which expression can be used to find the probability that a pet owner owns a cat or a dog?

- Ⓐ $P(C) + P(D) - P(C \cup D)$
- ● $P(C) + P(D) - P(C \cap D)$
- Ⓒ $P(C) + P(D) + P(C \cup D)$
- Ⓓ $P(C) + P(D) + P(C \cap D)$

Geometry
Spring 2025 Item Release

Question 27

Question and Scoring Guidelines

Question 27

There are 45 red marbles and 45 blue marbles in a bag. Santino draws 1 red marble from the bag and does not replace it. Then he draws 1 more marble from the bag.

Select a word or phrase for each blank box to describe the likelihood that the second marble Santino draws is blue.

Santino drawing a blue marble on the second draw is [ ] likely to happen than it was on the first draw because the two events are [ ]

Drop Down Choices

Santino drawing a blue marble on the second draw is

[ ] likely to happen than it was on the first

- ✔
- less
- more
- equally

draw because the two events are [ ]

- ✔
- dependent.
- independent.
- neither dependent nor independent.

Points Possible: 1

Content Cluster: Understand independence and conditional probability, and use them to interpret data.

Content Standard: Recognize and explain the concepts of conditional probability and independence in everyday language and everyday situations. *For example, compare the chance of having lung cancer if you are a smoker with the chance of being a smoker if you have lung cancer. (S.CP.5)*

Depth of Knowledge: Level 2

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student selects "more" and "dependent," providing evidence of the ability to recognize and explain the concepts of conditional probability and independence in everyday language.

Exemplar Response

- Santino drawing a blue marble on the second draw is **more** likely to happen than it was on the first draw because the two events are **dependent**.

Geometry
Spring 2025 Item Release

Question 27

Sample Responses

Sample Response: 1 point

There are 45 red marbles and 45 blue marbles in a bag. Santino draws 1 red marble from the bag and does not replace it. Then he draws 1 more marble from the bag.

Select a word or phrase for each blank box to describe the likelihood that the second marble Santino draws is blue.

Santino drawing a blue marble on the second draw is [more] likely to happen than it was on the first draw because the two events are [dependent.]

Notes on Scoring

This response receives full credit (1 point) because it correctly describes the likelihood that the second marble is blue.

The student understands that the probability of drawing a blue marble is calculated by dividing the number of blue marbles in the bag by the total number of marbles in the bag. For the first draw, the probability of drawing a blue marble is $\frac{45}{90}$ because there are 45 blue marbles and 90 total marbles in the bag. Since a red marble is selected on the first draw, there are still 45 blue marbles available for the second draw, but only 89 total marbles remain in the bag, making the probability of drawing a blue marble on the second draw $\frac{45}{89}$. The student recognizes that $\frac{45}{90} < \frac{45}{89}$, so drawing a blue marble on the second draw is more likely. The student understands that the events are dependent because the outcome of the first event changes the probability of the second event.

Sample Response: 0 points

There are 45 red marbles and 45 blue marbles in a bag. Santino draws 1 red marble from the bag and does not replace it. Then he draws 1 more marble from the bag.

Select a word or phrase for each blank box to describe the likelihood that the second marble Santino draws is blue.

Santino drawing a blue marble on the second draw is [equally] likely to happen than it was on the first draw because the two events are [independent.]

Notes on Scoring

This response receives no credit (0 points) because it does not correctly describe the likelihood that the second marble is blue.

The student may understand that the probability of drawing a blue marble is calculated by dividing the number of blue marbles in the bag by the total number of marbles in the bag. For the first draw, the probability of drawing a blue marble is $\frac{45}{90}$ because there are 45 blue marbles and 90 total marbles in the bag. The student may not realize that the total number of marbles for the second draw is only 89 because a red marble was removed, instead thinking that since the number of blue balls remains the same, the number of total marbles also remains the same. The student calculates the probability of drawing a blue marble as $\frac{45}{90}$ for both events and concludes the events are independent because the outcome of the first event does not change the probability of the second event.

Sample Response: 0 points

There are 45 red marbles and 45 blue marbles in a bag. Santino draws 1 red marble from the bag and does not replace it. Then he draws 1 more marble from the bag.

Select a word or phrase for each blank box to describe the likelihood that the second marble Santino draws is blue.

Santino drawing a blue marble on the second draw is [more] likely to happen than it was on the first draw because the two events are [independent.]

Notes on Scoring

This response receives no credit (0 points) because it does not correctly describe the likelihood that the second marble is blue.

The student may understand that the probability of drawing a blue marble is calculated by dividing the number of blue marbles in the bag by the total number of marbles in the bag. The student calculates $\frac{45}{90}$ as the probability of drawing a blue marble on draw 1 and $\frac{45}{89}$ as the probability of drawing a blue marble on draw 2. The student recognizes that $\frac{45}{90} < \frac{45}{89}$, so drawing a blue marble on the second draw is more likely. The student may confuse dependent and independent events and think the events are independent because the outcome of the first event changes the probability of the second event.

Geometry
Spring 2025 Item Release

Question 30

Question and Scoring Guidelines

Question 30

This item has **two** parts.

Carlos runs every morning. Every day after his morning run, he records the average temperature, in degrees Fahrenheit (°*F*), and whether or not he saw other runners.

- Carlos makes a table to compare the data during his last 30 runs.
- The average temperature was above 65°*F* more days than it was below 65°*F*.
- He did not see other runners on 5 days when the average temperature was below 65°*F*.
- He saw other runners twice as frequently when the average temperature was above 65°*F* than when it was below 65°*F*.

Part A. Complete the table to show a possible number of days each event occurred.

	Sees no runners	**Sees other runners**
Below 65°*F*	5	
Above 65°*F*		

Part B. Based on the data in the table, what is the probability that Carlos will see another runner given that the average temperature is below 65°*F*?



Points Possible: 2

Content Cluster: Understand independence and conditional probability, and use them to interpret data.

Content Standard: Construct and interpret two-way frequency tables of data when two categories are associated with each object being classified. Use the two-way table as a sample space to decide if events are independent and to approximate conditional probabilities. *For example, collect data from a random sample of students in your school on their favorite subject among math, science, and English. Estimate the probability that a randomly selected student from your school will favor science given that the student is in tenth grade. Do the same for other subjects and compare the results. (S.CP.4)*

Depth of Knowledge: Level 3

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (2 points), the student's response satisfies both the bullets below. For partial credit (1 point), the student's response satisfies one of the bullets below, including a correct response in Part B based on an incorrect response entered in Part A.

- The student correctly completes the two-way table, providing evidence of the ability to construct and interpret two-way frequency tables of data when two categories are associated with each object being classified.
- The student determines the conditional probability, providing evidence of the ability to determine the conditional probability from a two-way table.

Exemplar Response

2 points

Part A

	Sees no runners	Sees other runners
Below 65°F	5	7
Above 65°F	4	14

or

- any values such that the 4 numbers add to 30, the value in the bottom right corner is twice the value in the top right corner, and the sum of the values in the top row is less than the sum of the values in the bottom row.

Part B

- $\frac{7}{12}$
  or
- any fraction equivalent to $\frac{\textit{the number in the top right corner}}{\textit{the sum of the top row}}$

1 point

Part A

	Sees no Runners	Sees other runners
Below 65°F	5	9
Above 65°F	8	8

Part B

- $\frac{9}{14}$
  or
- any equivalent value

Geometry
Spring 2025 Item Release

Question 30

Sample Responses

Sample Response: 2 points

This item has **two** parts.

Carlos runs every morning. Every day after his morning run, he records the average temperature, in degrees Fahrenheit (°*F*), and whether or not he saw other runners.

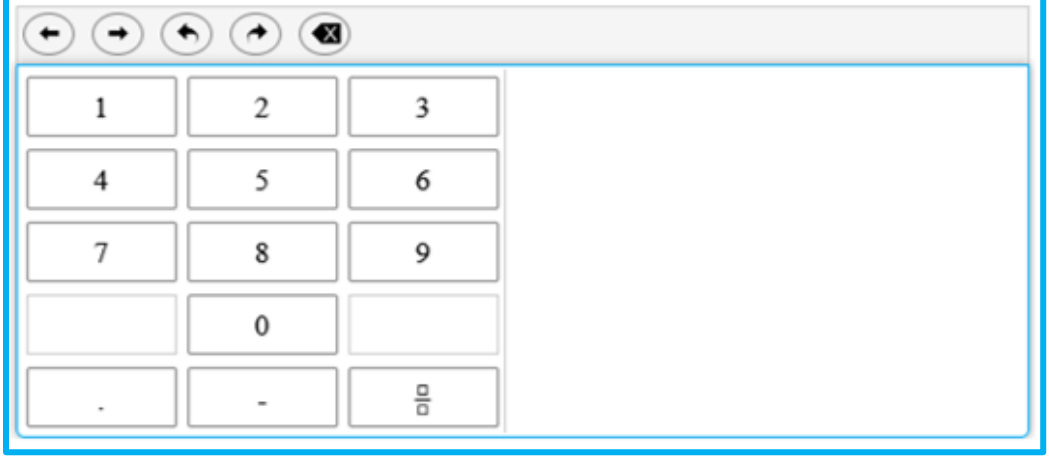
- Carlos makes a table to compare the data during his last 30 runs.
- The average temperature was above 65°*F* more days than it was below 65°*F*.
- He did not see other runners on 5 days when the average temperature was below 65°*F*.
- He saw other runners twice as frequently when the average temperature was above 65°*F* than when it was below 65°*F*.

Part A. Complete the table to show a possible number of days each event occurred.

	Sees no runners	Sees other runners
Below 65°*F*	5	8
Above 65°*F*	1	16

Part B. Based on the data in the table, what is the probability that Carlos will see another runner given that the average temperature is below 65°*F*?

$\frac{8}{13}$



Notes on Scoring

This response receives full credit (2 points) because it identifies possible values for the two-way table and the corresponding probability.

The student may start with the information that the average temperature was above 65°*F* more days than it was below 65°*F* and enters 16 for **sees other runners** and **above 65°*F*** because 16 is more than half of the days. Then, because Carlos sees other runners twice as many times when the temperature is above 65°*F* than when it is below 65°*F*, the student may divide 16 by 2, entering 8 for **sees other runners** and **below 65°*F***. The student may then add the three numbers in the table 16 + 8 + 5 = 29 and enters 1 for **sees no runners** and **above 65°*F*** so all four values add up to 30.

To calculate the probability that Carlos will see another runner given that the average temperature is below 65°*F*, the student divides the number of days Carlos sees other runners when the average temperature is below 65°*F*, 8, by the total number of days the average temperature is below 65°*F*, 13. So, the probability is $\frac{8}{13}$.

Sample Response: 2 points

This item has **two** parts.

Carlos runs every morning. Every day after his morning run, he records the average temperature, in degrees Fahrenheit (°*F*), and whether or not he saw other runners.

- Carlos makes a table to compare the data during his last 30 runs.
- The average temperature was above 65°*F* more days than it was below 65°*F*.
- He did not see other runners on 5 days when the average temperature was below 65°*F*.
- He saw other runners twice as frequently when the average temperature was above 65°*F* than when it was below 65°*F*.

Part A. Complete the table to show a possible number of days each event occurred.

	Sees no runners	Sees other runners
Below 65°*F*	5	5
Above 65°*F*	10	10

Part B. Based on the data in the table, what is the probability that Carlos will see another runner given that the average temperature is below 65°*F*?

$\frac{1}{2}$



Notes on Scoring

This response receives full credit (2 points) because it identifies possible values for the two-way table and the corresponding probability.

The student may start with the information that the average temperature was above 65°*F* more days than it was below 65°*F* and enters 5 for **sees other runners** and **below 65°*F*** because 5 + 5 is less than half the days. Then, the student may multiply 5 by 2 because Carlos sees other runners twice as many times when the temperature is above 65°*F* than when it is below 65°*F*, entering 10 for **sees other runners** and **above 65°*F***. The student may then add the three numbers in the table 10 + 5 + 5 = 20 and enters 10 for **sees no runners** and **above 65°*F***, so all four values add up to 30.

To calculate the probability that Carlos will see another runner given that the average temperature is below 65°*F*, the student divides the number of days Carlos sees other runners when the average temperature is below 65°*F*, 5, by the total number of days the average temperature is below 65°*F*, 5 + 5, or 10. So, the probability is $\frac{5}{10}$ or $\frac{1}{2}$.

Sample Response: 1 point

This item has **two** parts.

Carlos runs every morning. Every day after his morning run, he records the average temperature, in degrees Fahrenheit (°*F*), and whether or not he saw other runners.

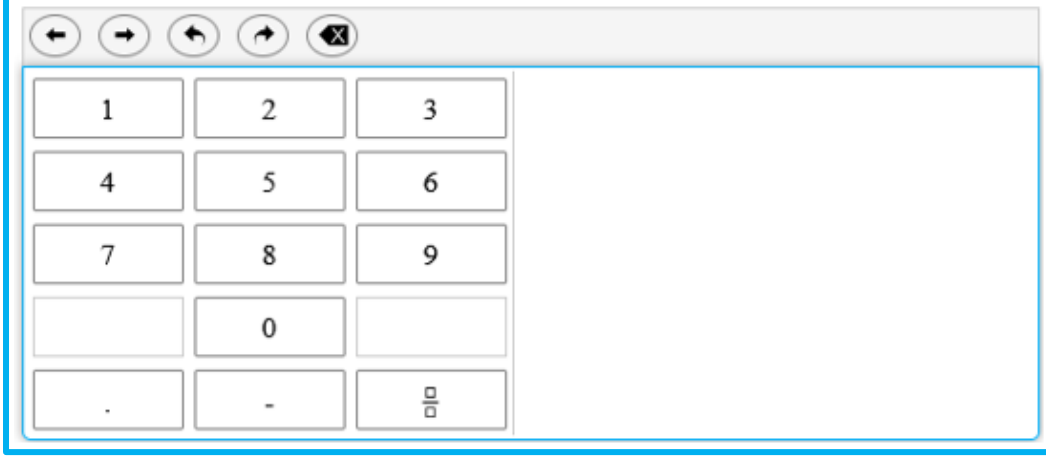
- Carlos makes a table to compare the data during his last 30 runs.
- The average temperature was above 65°*F* more days than it was below 65°*F*.
- He did not see other runners on 5 days when the average temperature was below 65°*F*.
- He saw other runners twice as frequently when the average temperature was above 65°*F* than when it was below 65°*F*.

Part A. Complete the table to show a possible number of days each event occurred.

	Sees no runners	**Sees other runners**
Below 65°*F*	5	8
Above 65°*F*	1	16

Part B. Based on the data in the table, what is the probability that Carlos will see another runner given that the average temperature is below 65°*F*?

$\frac{8}{30}$



Notes on Scoring

This response receives partial credit (1 point) because it identifies possible values for the two-way table but not a correct probability.

The student may start with the information that the average temperature was above 65°*F* more days than it was below 65°*F* and enters 16 for **sees other runners** and **above 65°*F*** because 16 is more than half of the days. Then, because Carlos sees other runners twice as many times when the temperature is above 65°*F* than when it is below 65°*F*, the student may divide 16 by 2, entering 8 for **sees other runners** and **below 65°*F***. The student may then add the three numbers in the table 16 + 8 + 5 = 29 and enters 1 for **sees no runners** and **above 65°*F***, so all four values add up to 30.

When calculating the probability that Carlos will see another runner given that the average temperature is below 65°*F*, the student may incorrectly use the total number of days for the denominator instead of the number of days the temperature is below 65°*F*, calculating $\frac{8}{30}$ instead of $\frac{8}{13}$.

Sample Response: 1 point

This item has **two** parts.

Carlos runs every morning. Every day after his morning run, he records the average temperature, in degrees Fahrenheit (°*F*), and whether or not he saw other runners.

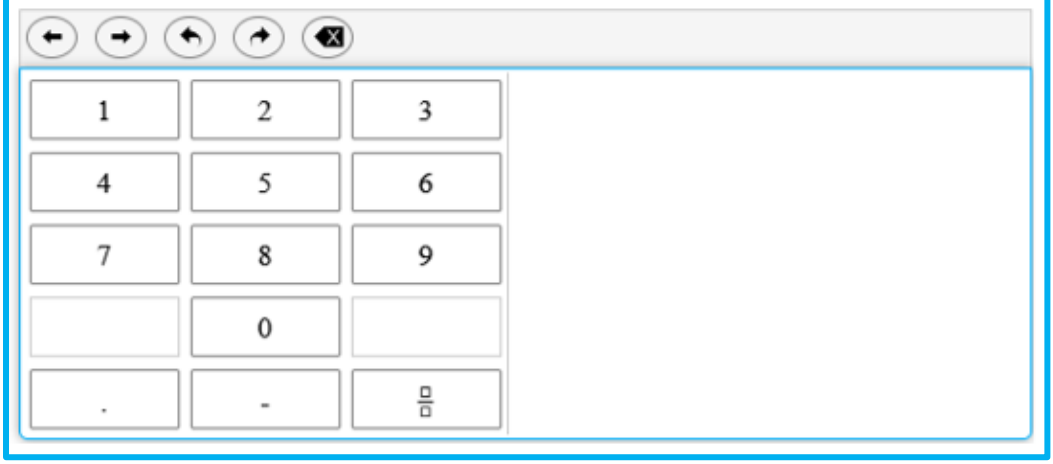
- Carlos makes a table to compare the data during his last 30 runs.
- The average temperature was above 65°*F* more days than it was below 65°*F*.
- He did not see other runners on 5 days when the average temperature was below 65°*F*.
- He saw other runners twice as frequently when the average temperature was above 65°*F* than when it was below 65°*F*.

Part A. Complete the table to show a possible number of days each event occurred.

	Sees no runners	Sees other runners
Below 65°*F*	5	10
Above 65°*F*	5	10

Part B. Based on the data in the table, what is the probability that Carlos will see another runner given that the average temperature is below 65°*F*?

$\frac{10}{15}$



Notes on Scoring

This response receives partial credit (1 point) because it identifies incorrect values for the two-way table and a correct probability.

The student may misunderstand which value should be "twice as much" and may think Carlos **sees other runners** twice as many times as he **sees no runners** regardless of the temperature. The student then enters 10 for **sees other runners** and **below 65°F** because $10 = 2 \times 5$. Since 15 days were below 65°*F*, the student puts the remaining 15 days in the above 65°*F* row. The student uses 5 for **sees no runners** and 10 for **sees other runners** because of the misconception that Carlos **sees other runners** twice as many times as he **sees no runners**.

Even though the student enters incorrect values in the table, the student calculates the correct probability that Carlos will see another runner given that the average temperature is below 65°*F* based on that table. The student divides the number of days Carlos sees other runners when the average temperature is below 65°*F*, 10, by the total number of days the average temperature is below 65°*F*, 15. So, the probability is $\frac{10}{15}$.

Sample Response: 0 points

This item has **two** parts.

Carlos runs every morning. Every day after his morning run, he records the average temperature, in degrees Fahrenheit (°*F*), and whether or not he saw other runners.

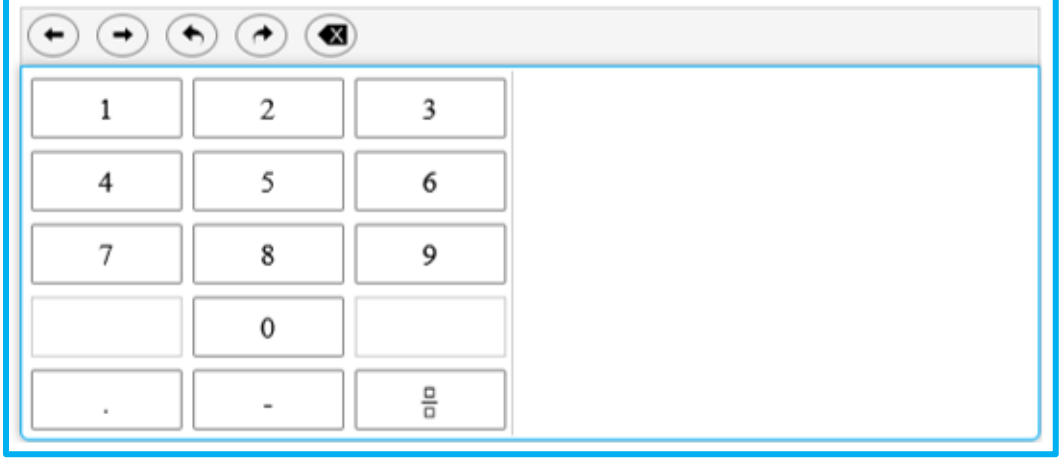
- Carlos makes a table to compare the data during his last 30 runs.
- The average temperature was above 65°*F* more days than it was below 65°*F*.
- He did not see other runners on 5 days when the average temperature was below 65°*F*.
- He saw other runners twice as frequently when the average temperature was above 65°*F* than when it was below 65°*F*.

Part A. Complete the table to show a possible number of days each event occurred.

	Sees no runners	Sees other runners
Below 65°*F*	5	10
Above 65°*F*	5	10

Part B. Based on the data in the table, what is the probability that Carlos will see another runner given that the average temperature is below 65°*F*?

$\frac{10}{30}$



Notes on Scoring

This response receives no credit (0 points) because it identifies incorrect values for the two-way table and an incorrect probability.

The student may misunderstand which value should be "twice as much" and may think Carlos **sees other runners** twice as many times as he **sees no runners** regardless of the temperature. The student then enters 10 for **sees other runners** and **below 65°*F*** because $10 = 2 \times 5$. Since 15 days were below 65°*F*, the student puts the remaining 15 days in the above 65°*F* row. The student uses 5 for **sees no runners** and 10 for **sees other runners** because of the misconception that Carlos **sees other runners** twice as many times as he **sees no runners**.

When calculating the probability that Carlos will see another runner given that the average temperature is below 65°*F*, the student may incorrectly use the total number of days for the denominator instead of the number of days the temperature is below 65°*F*, calculating $\frac{10}{30}$ instead of $\frac{10}{15}$.

Sample Response: 0 points

This item has **two** parts.

Carlos runs every morning. Every day after his morning run, he records the average temperature, in degrees Fahrenheit (°*F*), and whether or not he saw other runners.

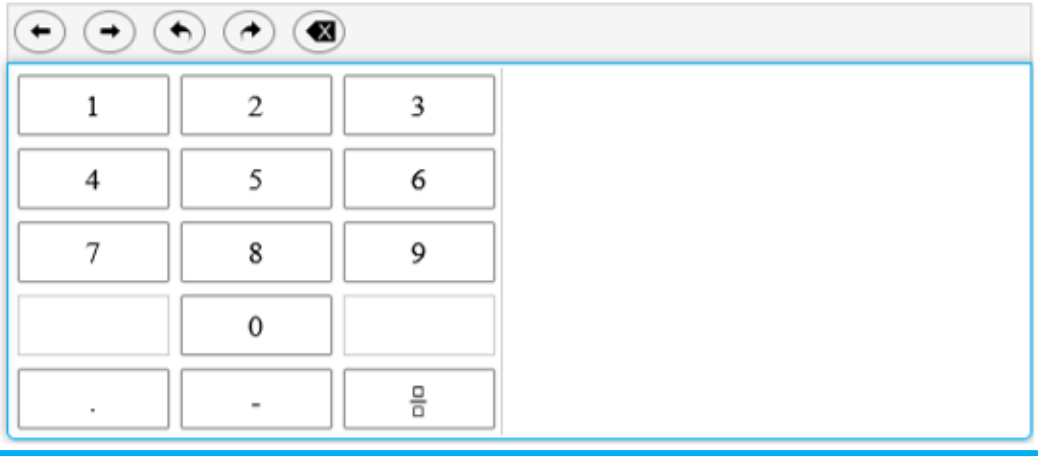
- Carlos makes a table to compare the data during his last 30 runs.
- The average temperature was above 65°*F* more days than it was below 65°*F*.
- He did not see other runners on 5 days when the average temperature was below 65°*F*.
- He saw other runners twice as frequently when the average temperature was above 65°*F* than when it was below 65°*F*.

Part A. Complete the table to show a possible number of days each event occurred.

	Sees no runners	Sees other runners
Below 65°*F*	5	8
Above 65°*F*	10	16

Part B. Based on the data in the table, what is the probability that Carlos will see another runner given that the average temperature is below 65°*F*?

$\frac{1}{2}$



Notes on Scoring

This response receives no credit (0 points) because it identifies incorrect values for the two-way table and an incorrect probability.

The student may start with the information that the average temperature was above 65°*F* more days than it was below 65°*F* and enters 16 for **sees other runners** and **above 65°*F*** because 16 is more than half of the days. Then, because Carlos sees other runners twice as many times when the temperature is above 65°*F* than when it is below 65°*F*, the student may divide 16 by 2, entering 8 for **sees other runners** and **below 65°*F***. The student may think Carlos also sees other runners twice as many times when the temperature is above 65°*F* than when it is below 65°*F*. The student then enters 10 for **sees no runners** and **above 65°*F*** because 10 = 2 × 5. The student may not add all the numbers and does not realize that the table represents 39 days instead of 30.

When calculating the probability that Carlos will see another runner given that the average temperature is below 65°*F*, the student may incorrectly use the number of days Carlos will see another runner given that the average temperature is below 65°*F* for the denominator, calculating $\frac{8}{16}$, which is equivalent to $\frac{1}{2}$.

Geometry
Spring 2025 Item Release

Question 33

Question and Scoring Guidelines

Question 33

A bag has pink and black bracelets. Jeremy picks a black bracelet from the bag and does not replace it.

- Event X: Picking a black bracelet.
- Event Y: Picking a pink bracelet.

Jeremy picks a second bracelet from the bag.

Which expression describes the probability that the second bracelet is pink?

Ⓐ $P(Y \mid X)$

Ⓑ $P(X \mid Y)$

Ⓒ $P(X \cup Y)$

Ⓓ $P(X \cap Y)$

Points Possible: 1

Content Cluster: Use the rules of probability to compute probabilities of compound events in a uniform probability model.

Content Standard: Find the conditional probability of A given B as the fraction of B's outcomes that also belong to A, and interpret the answer in terms of the model. *(S.CP.6)*

Depth of Knowledge: Level 1

Modeling and Reasoning: Yes

Scoring Guidelines

Rationale for Option A: **Key** – The student recognizes this situation represents a conditional probability and that event *Y* is dependent on event *X*. The student identifies the probability of event *Y*, choosing a pink bracelet, given event *X*, choosing a black bracelet.

Rationale for Option B: This is incorrect. The student may recognize this situation represents a conditional probability but confuses which event was dependent on the other, thinking the situation is represented by the probability of event *X*, choosing a black bracelet, given event *Y*, choosing a pink bracelet.

Rationale for Option C: This is incorrect. The student may confuse union notation with notation for conditional probability.

Rationale for Option D: This is incorrect. The student may confuse intersection notation with notation for conditional probability.

Sample Response: 1 point

A bag has pink and black bracelets. Jeremy picks a black bracelet from the bag and does not replace it.

- Event *X*: Picking a black bracelet.
- Event *Y*: Picking a pink bracelet.

Jeremy picks a second bracelet from the bag.

Which expression describes the probability that the second bracelet is pink?

● $P(Y \mid X)$

Ⓑ $P(X \mid Y)$

Ⓒ $P(X \cup Y)$

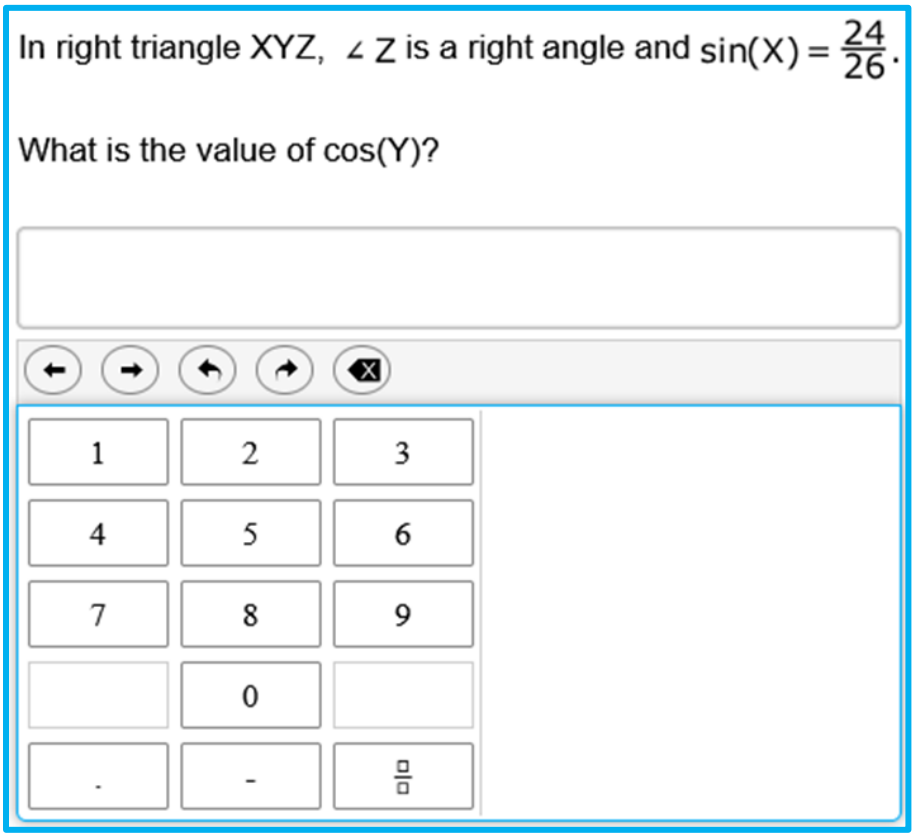
Ⓓ $P(X \cap Y)$

Geometry
Spring 2025 Item Release

Question 35

Question and Scoring Guidelines

Question 35



Points Possible: 1

Content Cluster: Define trigonometric ratios and solve problems involving right triangles.

Content Standard: Explain and use the relationship between the sine and cosine of complementary angles. *(G.SRT.7)*

Depth of Knowledge: Level 1

Modeling and Reasoning: No

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student enters $\frac{24}{26}$, providing evidence of the ability to use the relationship between the sine and cosine of complementary angles.

Exemplar Response

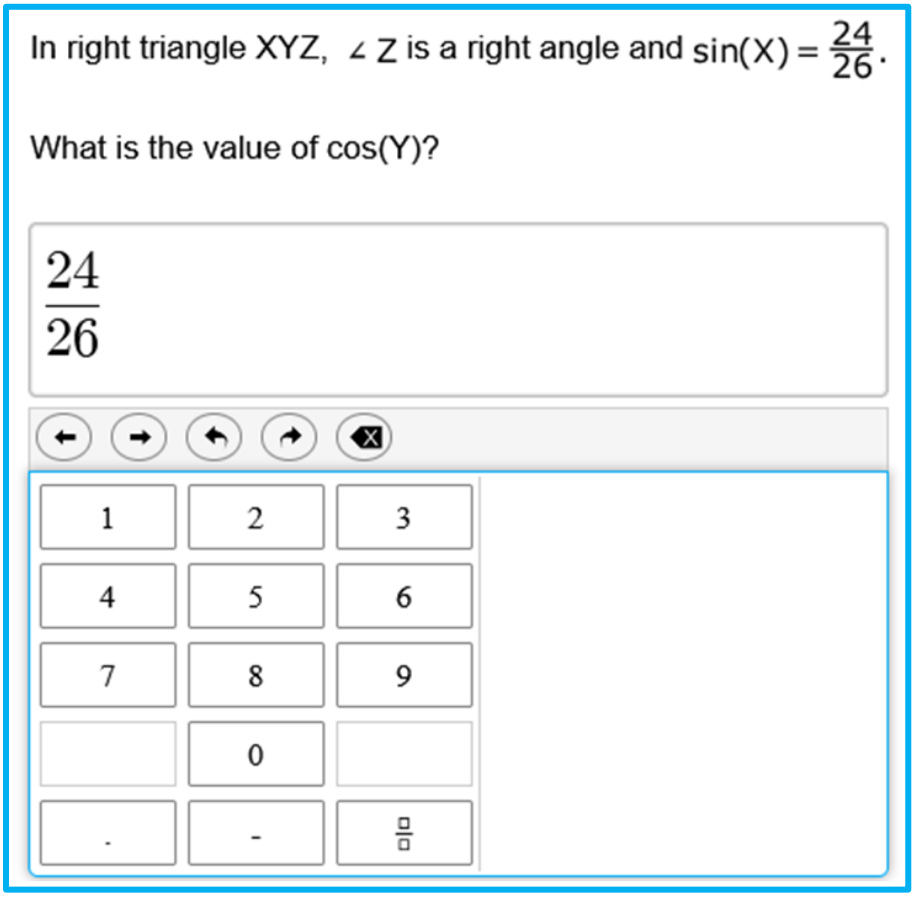
- $\frac{24}{26}$
  or
- any equivalent value

Geometry
Spring 2025 Item Release

Question 35

Sample Responses

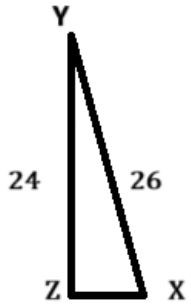
Sample Response: 1 point



Notes on Scoring

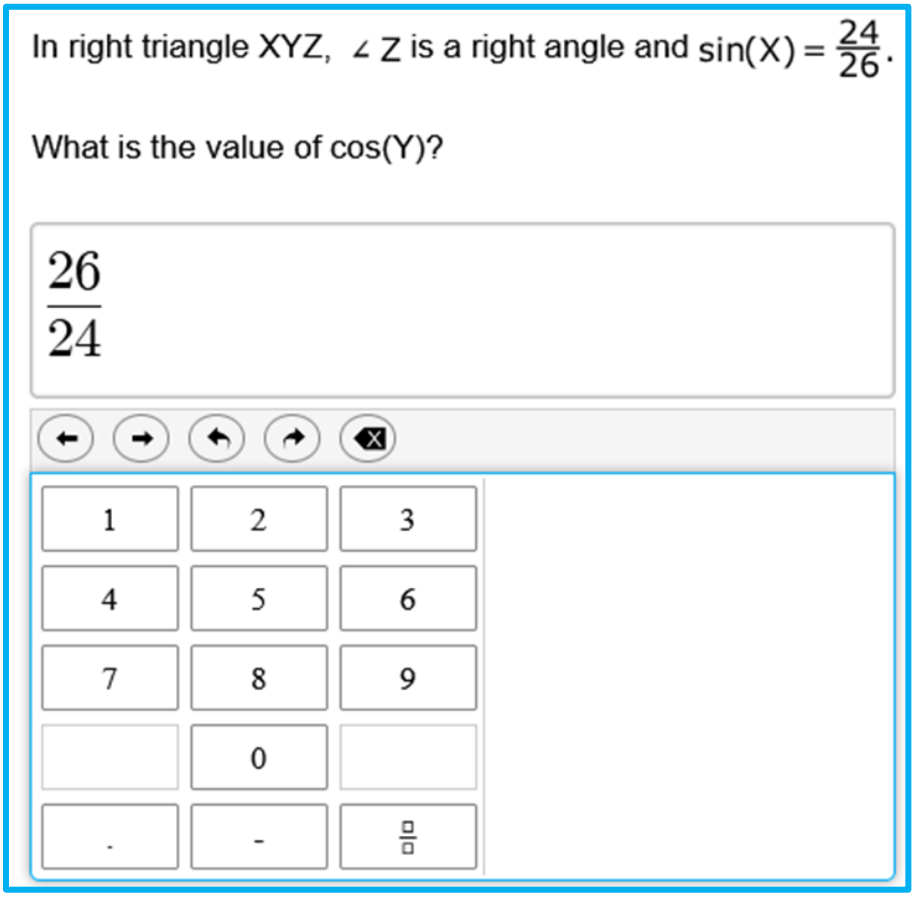
This response receives full credit (1 point) because it provides the correct value of cos(Y).

The student may sketch a right triangle such that the sine of angle X, $\frac{\text{the length opposite angle X}}{\text{the hypotenuse}}$, is $\frac{24}{26}$.



The student may then use the sketch to determine that the cosine of angle Y, $\frac{\text{the length adjacent to angle Y}}{\text{the hypotenuse}}$, is also $\frac{24}{26}$.

Sample Response: 0 points

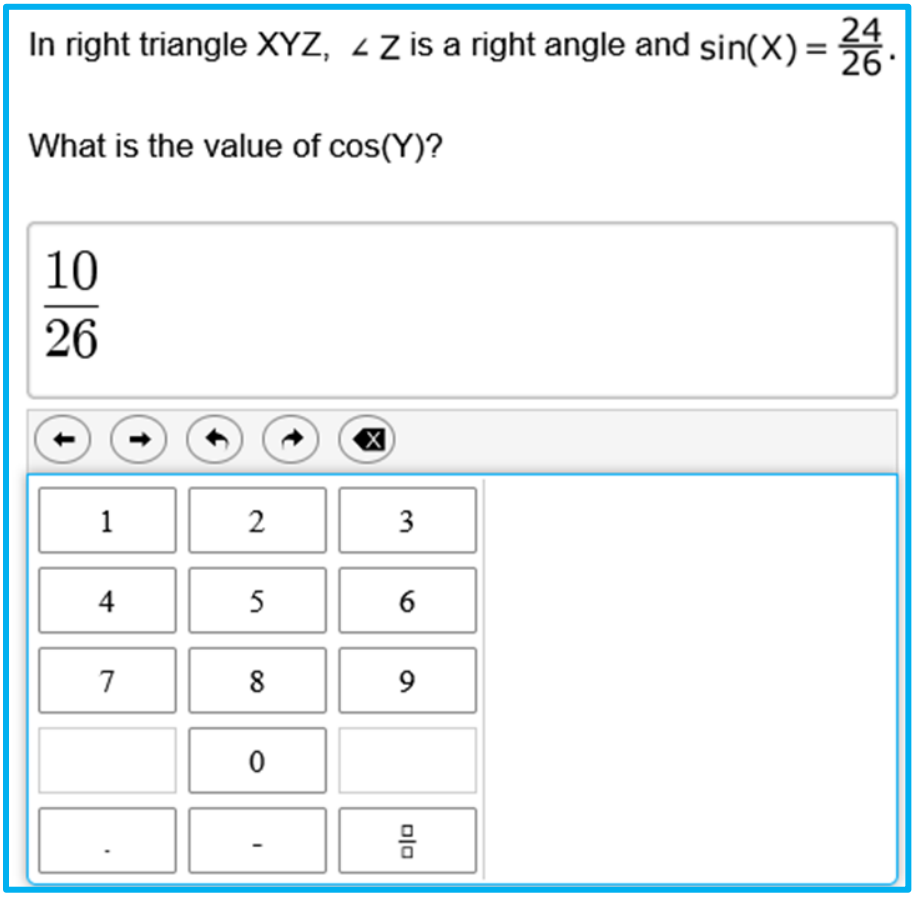


Notes on Scoring

This response receives no credit (0 points) because it does not provide the correct value of cos(Y).

The student may think that cos(Y) is equal to the reciprocal of sin(X), or $\frac{26}{24}$.

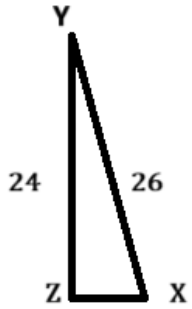
Sample Response: 0 points



Notes on Scoring

This response receives no credit (0 points) because it does not provide the correct value of cos(Y).

The student may sketch a right triangle such that the sine of angle X, $\frac{\text{the length opposite angle X}}{\text{the hypotenuse}}$, is $\frac{24}{26}$.



The student may then use the Pythagorean Theorem to calculate XZ as 10 units.

$$XZ^2 + 24^2 = 26^2$$

$$XZ^2 + 576 = 676$$

$$XZ^2 = 676 - 576$$

$$XZ^2 = 100$$

$$XZ = 10$$

The student may confuse cos(Y) with cos(X) and calculates $\frac{\text{the length adjacent to angle X}}{\text{the hypotenuse}}$, or $\frac{10}{26}$.

Geometry
Spring 2025 Item Release

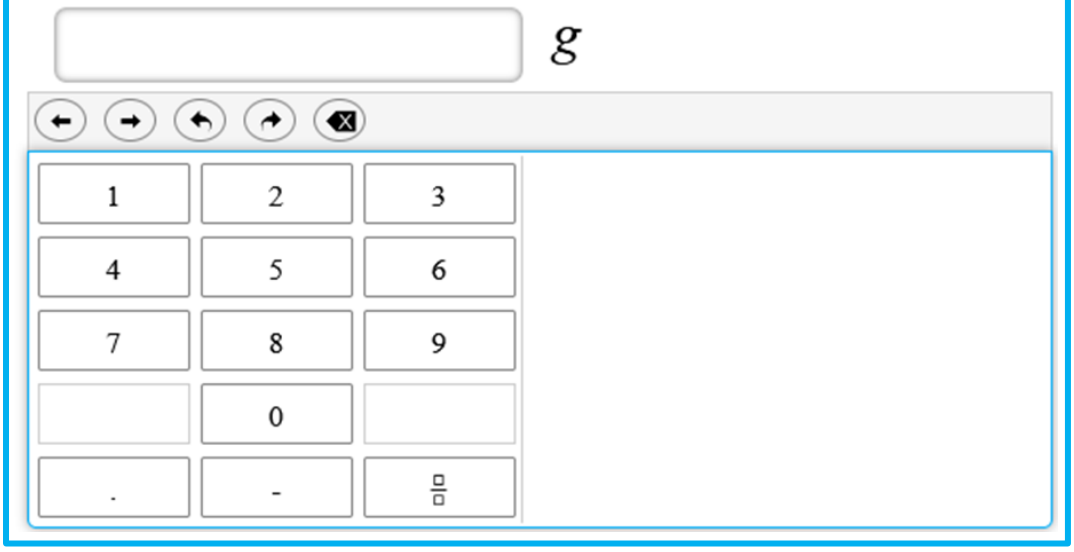
Question 37

Question and Scoring Guidelines

Question 37

The density of gold is 19.32 grams (g) per cubic centimeter (cm). A solid rectangular block of gold with dimensions 5 cm by 9 cm by 21 cm is placed on a scale.

What is the mass, in grams, of the solid block of gold?



Points Possible: 1

Content Cluster: Apply geometric concepts in modeling situations.

Content Standard: Apply concepts of density based on area and volume in modeling situations (e.g., persons per square mile, BTUs per cubic foot). *(G.MG.2)*

Depth of Knowledge: Level 2

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student calculates the mass of gold as 18257.4 g, providing evidence of understanding how to apply the concepts of density based on volume.

Exemplar Response

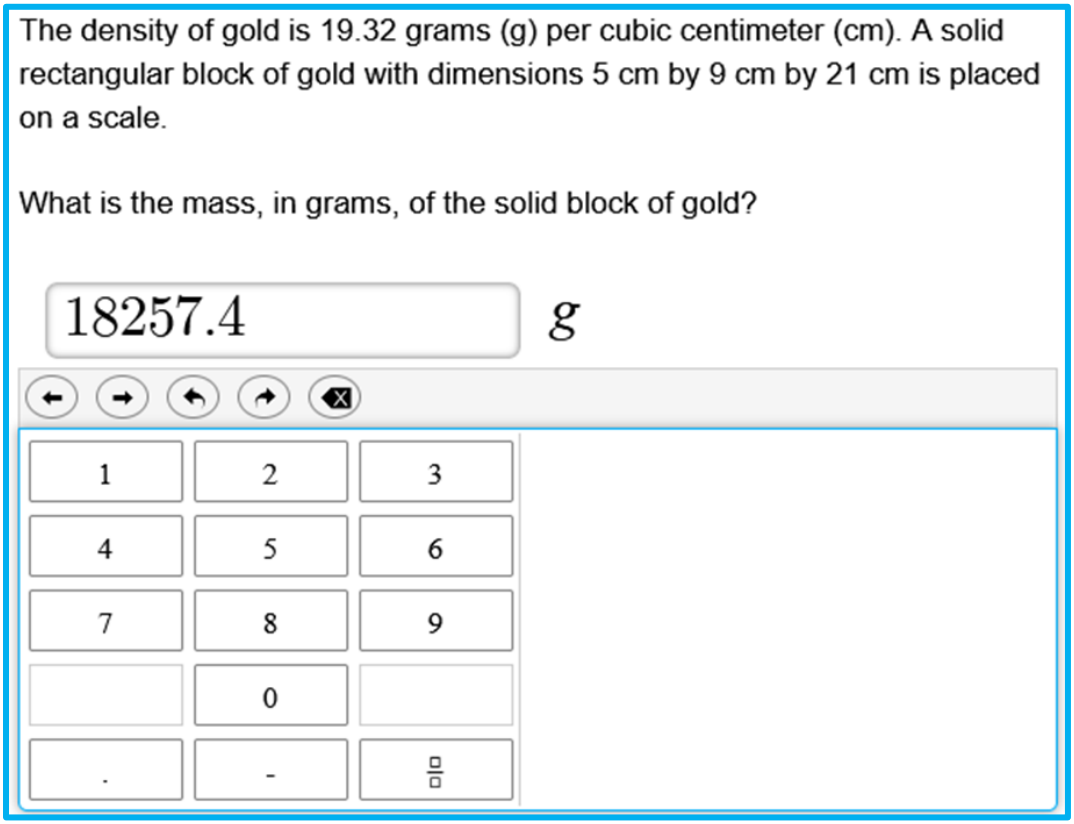
- 18257.4 g
  or
- any equivalent value

Geometry
Spring 2025 Item Release

Question 37

Sample Responses

Sample Response: 1 point



Notes on Scoring

This response receives full credit (1 point) because it identifies the correct mass of the solid block of gold.

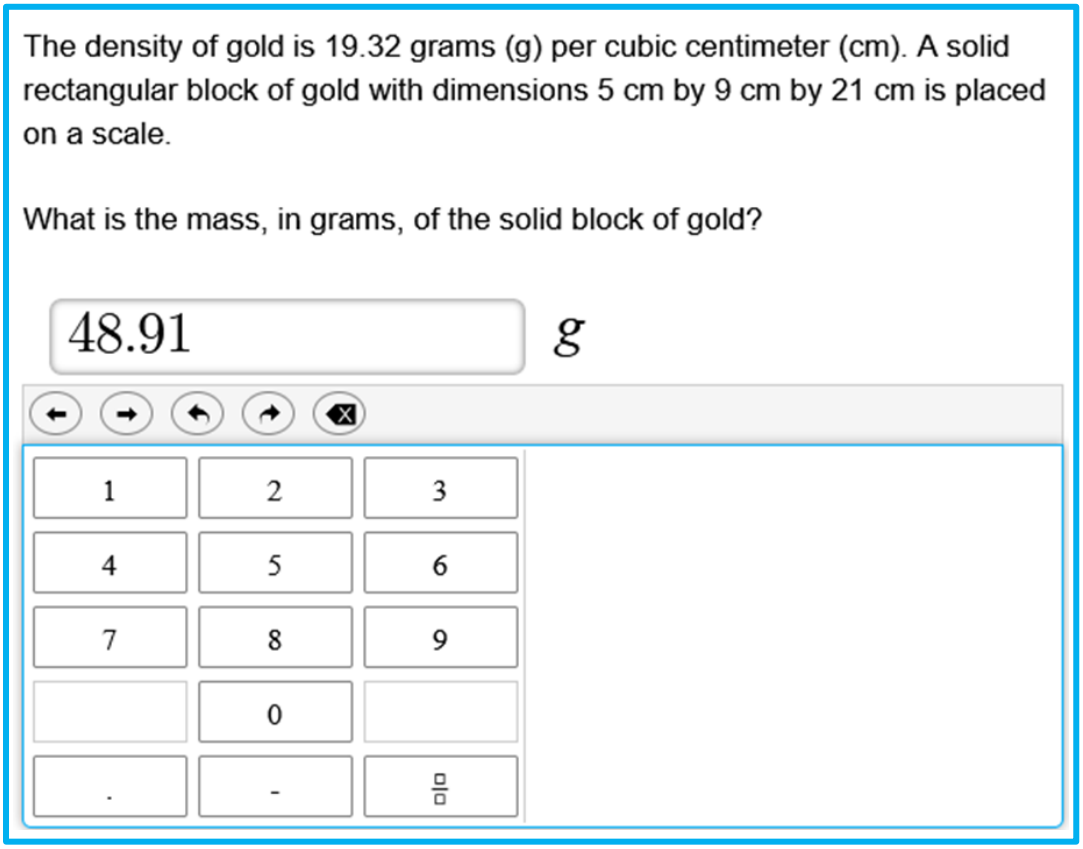
The student calculates the volume of the gold, in cubic centimeters, by multiplying the three dimensions.

$$5 \times 9 \times 21 = 945$$

The student calculates the mass of the gold by multiplying the volume by the density.

$$945\ cm^3 \times \frac{19.32\ g}{cm^3} = 18257.4\ g$$

Sample Response: 0 points



Notes on Scoring

This response receives no credit (0 points) because it does not identify the correct mass of the solid block of gold.

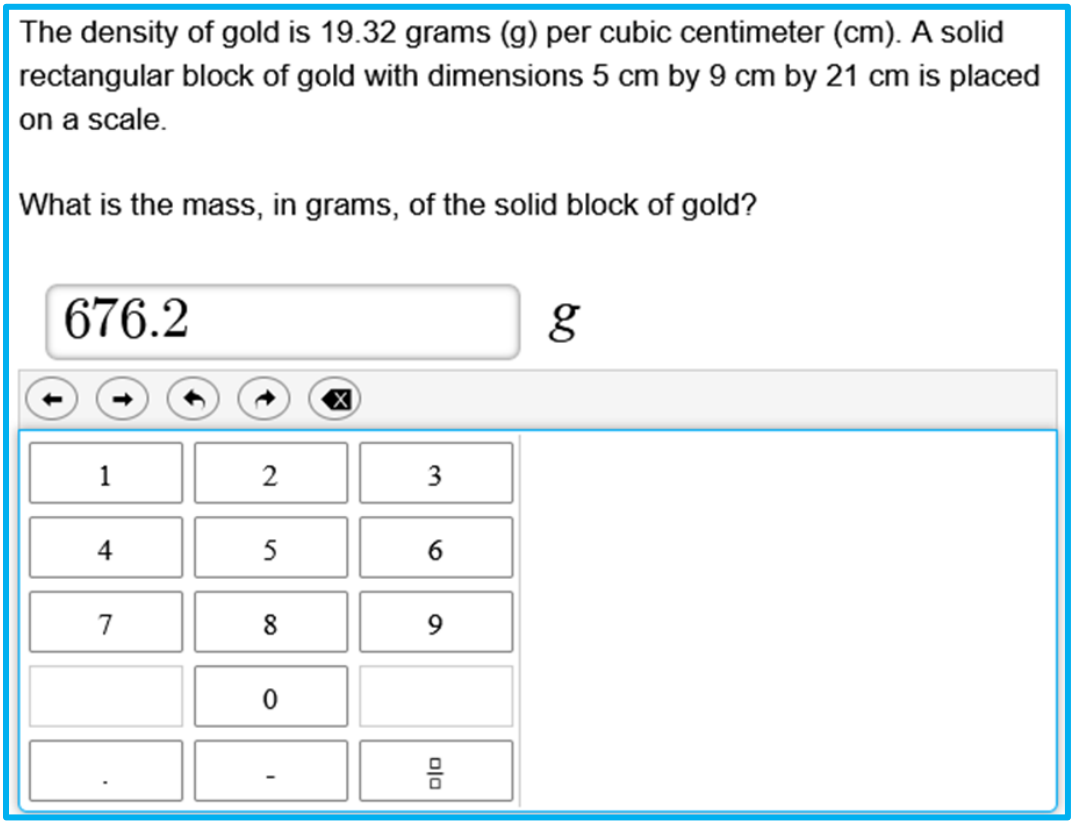
The student may calculate the volume of the gold, in cubic centimeters, by multiplying the three dimensions.

$$5 \times 9 \times 21 = 945$$

The student then calculates the mass of the gold by dividing the volume by the density, instead of multiplying, not realizing the units would be $\frac{cm^6}{g}$ instead of g.

$$945\ cm^3 \div \frac{19.32\ g}{cm^3} = 48.91\ g$$

Sample Response: 0 points



Notes on Scoring

This response receives no credit (0 points) because it does not identify the correct mass of the solid block of gold.

The student may calculate the volume of the gold, in cubic centimeters, by adding the three dimensions instead of multiplying.

$$5 + 9 + 21 = 35$$

The student then calculates the mass of the gold by multiplying the density by the volume.

$$\frac{19.32\ g}{cm^3} \times 35 m^3 = 676.2\ g$$

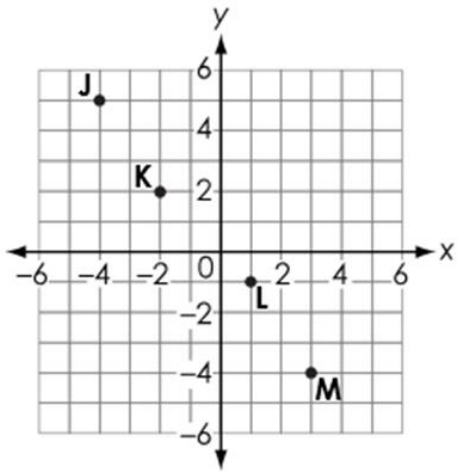
Geometry
Spring 2025 Item Release

Question 41

Question and Scoring Guidelines

Question 41

Points J, K, L, and M are shown on the coordinate plane.



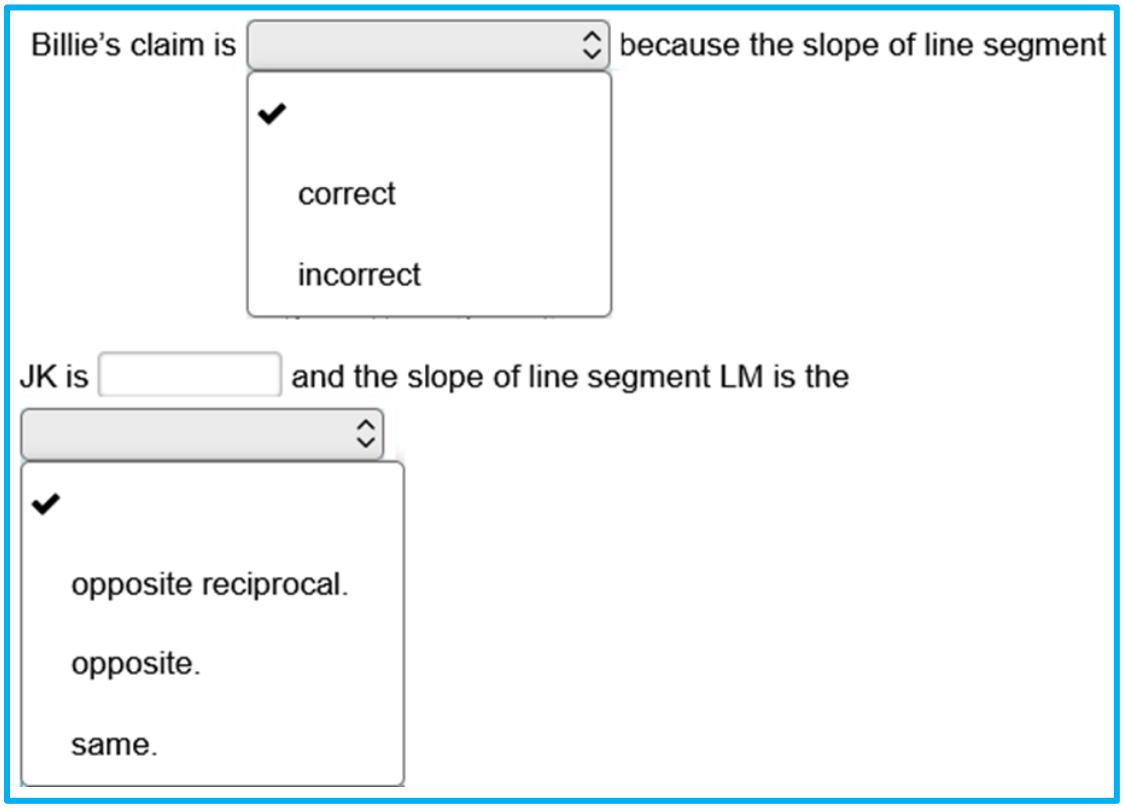
Billie claims line segment JK is parallel to line segment LM.

Complete the sentence describing Billie's claim.

Billie's claim is [ ] because the slope of line segment JK is [ ] and the slope of line segment LM is the [ ]

1	2	3
4	5	6
7	8	9
	0	
.	-	

Drop Down Choices



Points Possible: 1

Content Cluster: Use coordinates to prove simple geometric theorems algebraically and to verify specific geometric statements.

Content Standard: Justify the slope criteria for parallel and perpendicular lines, and use them to solve geometric problems, e.g., find the equation of a line parallel or perpendicular to a given line that passes through a given point. *(G.GPE.5)*

Depth of Knowledge: Level 2

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student selects "correct" and "same" and enters $-\frac{3}{2}$, providing evidence of the ability to justify the slope criteria for parallel lines and use it to solve geometric problems.

Exemplar Response

- Billie's claim is **correct** because the slope of line segment JK is $-\mathbf{\frac{3}{2}}$ and the slope of line segment LM is the **same.**
  or
- any equivalent value of $-\frac{3}{2}$

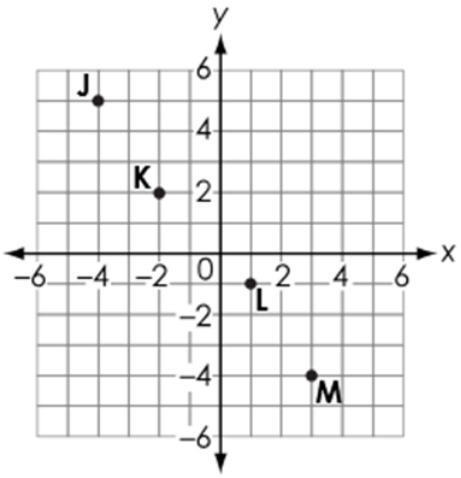
Geometry
Spring 2025 Item Release

Question 41

Sample Responses

Sample Response: 1 point

Points J, K, L, and M are shown on the coordinate plane.



Billie claims line segment JK is parallel to line segment LM.

Complete the sentence describing Billie's claim.

Billie's claim is [correct] because the slope of line segment JK is $-\frac{3}{2}$ and the slope of line segment LM is the [same.]

1 2 3
4 5 6
7 8 9
0
. -

Notes on Scoring

This response receives full credit (1 point) because it correctly describes the claim.

The student may recognize that they need to find the slope of line segments JK and LM. The student uses the expression $\frac{y_2-y_1}{x_2-x_1}$ to calculate the slope.

For line segment JK, the student finds the slope of the line using the points for J, (–4, 5), and K, (–2, 2). Point J can be used for (x_1 ,y_1), and point K can be used for (x_2 ,y_2). The student substitutes in the values and determines the slope is $-\frac{3}{2}$.

$$\frac{y_2-y_1}{x_2-x_1}=\frac{2-5}{-2-(-4)}$$

$$=\frac{-3}{2}$$

For line segment LM, the student calculates the slope of the line using the points for L, (1, –1), and M, (3, –4). Point L can be used for (x_1 ,y_1), and point M can be used for (x_2 ,y_2). The student substitutes in the values and determines the slope is $-\frac{3}{2}$.

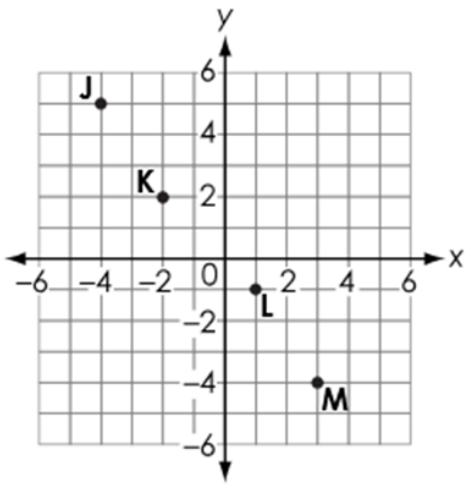
$$\frac{y_2-y_1}{x_2-x_1}=\frac{-4-(-1)}{3-1}$$

$$=\frac{-3}{2}$$

The slope for both line segments is the same, and all four points are not on the same line, so the two line segments are parallel. This means that Billie's claim is correct.

Sample Response: 0 points

Points J, K, L, and M are shown on the coordinate plane.



Billie claims line segment JK is parallel to line segment LM.

Complete the sentence describing Billie's claim.

Billie's claim is correct because the slope of line segment JK is $-\frac{3}{2}$ and the slope of line segment LM is the opposite reciprocal.



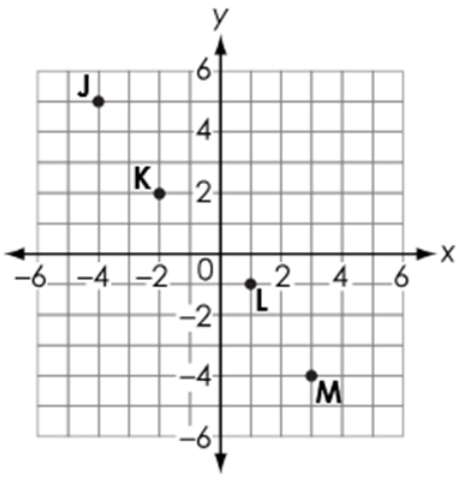
Notes on Scoring

This response receives no credit (0 points) because it does not correctly describe the claim.

The student may visually recognize that the two line segments look parallel and identify that Billie's claim is true without calculating the slope of LM. The student calculates the slope of line segment JK as $-\frac{3}{2}$ but may confuse the slope of perpendicular lines with the slope of parallel lines, thinking the slope of LM would be $\frac{2}{3}$, the opposite of the reciprocal for the slope of JK.

Sample Response: 0 points

Points J, K, L, and M are shown on the coordinate plane.



Billie claims line segment JK is parallel to line segment LM.

Complete the sentence describing Billie's claim.

Billie's claim is incorrect because the slope of line segment JK is $-\frac{3}{2}$ and the slope of line segment LM is the opposite.



Notes on Scoring

This response receives no credit (0 points) because it does not correctly describe the claim.

The student may recognize that they need to find the slope of line segments JK and LM. The student uses the expression $\frac{y_2-y_1}{x_2-x_1}$ to calculate the slope.

The student correctly calculates the slope of JK as $-\frac{3}{2}$. The student may incorrectly place the values from the points into $\frac{y_2-y_1}{x_2-x_1}$ when calculating the slope of LM, always putting the larger value first.

$\frac{y_2-y_1}{x_2-x_1} = \frac{-1-(-4)}{3-1}$

$= \frac{3}{2}$.

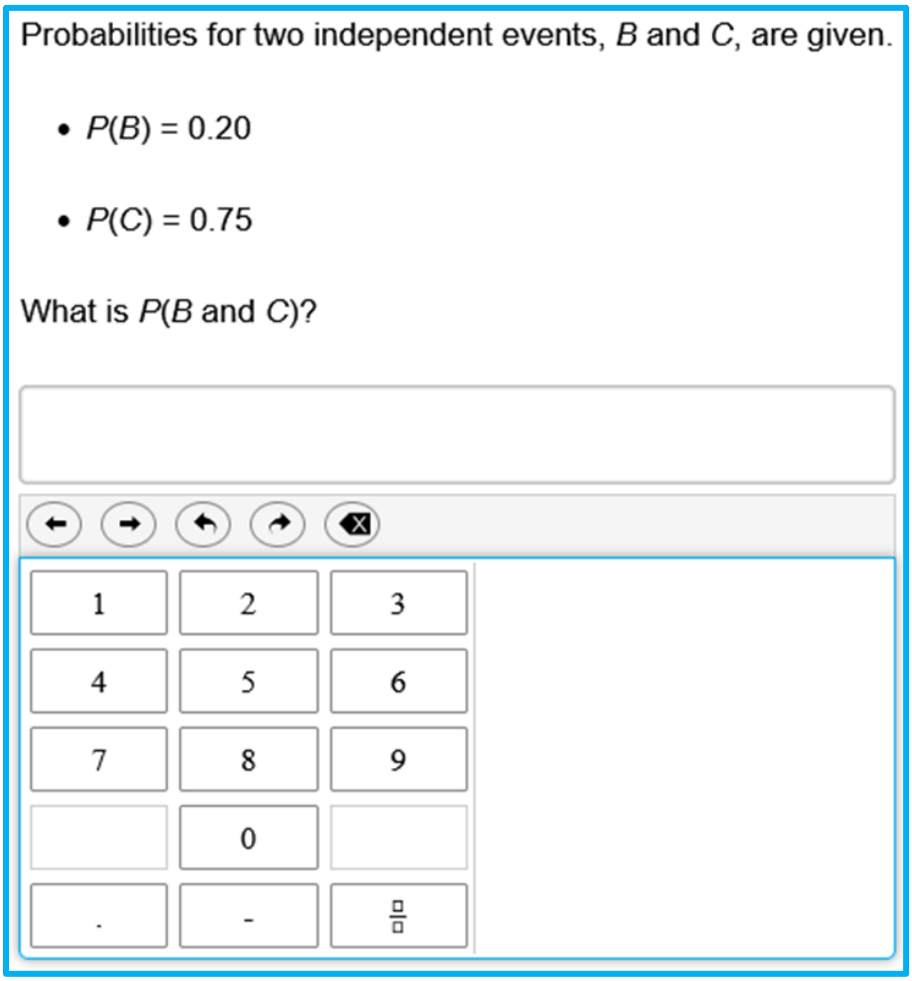
The student may incorrectly identify that the lines are not parallel because the slopes are opposites, which would mean that Billie's claim is incorrect.

Geometry
Spring 2025 Item Release

Question 42

Question and Scoring Guidelines

Question 42



Points Possible: 1

Content Cluster: Understand independence and conditional probability, and use them to interpret data.

Content Standard: Understand that two events A and B are independent if and only if the probability of A and B occurring together is the product of their probabilities, and use this characterization to determine if they are independent. *(S.CP.2)*

Depth of Knowledge: Level 1

Modeling and Reasoning: No

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student identifies the probability as 0.15, providing evidence of the ability to recognize that if two events are independent, the probability of them occurring together is the product of their probabilities.

Exemplar Response

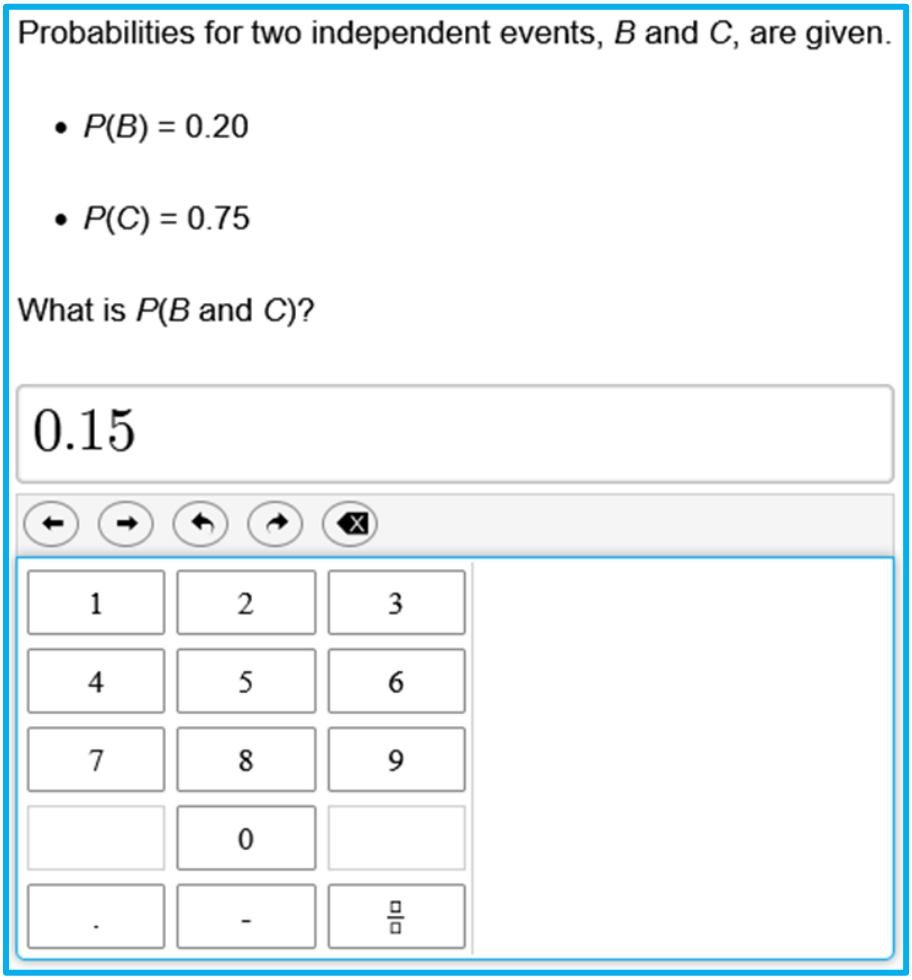
- 0.15
  or
- any equivalent value

Geometry
Spring 2025 Item Release

Question 42

Sample Responses

Sample Response: 1 point



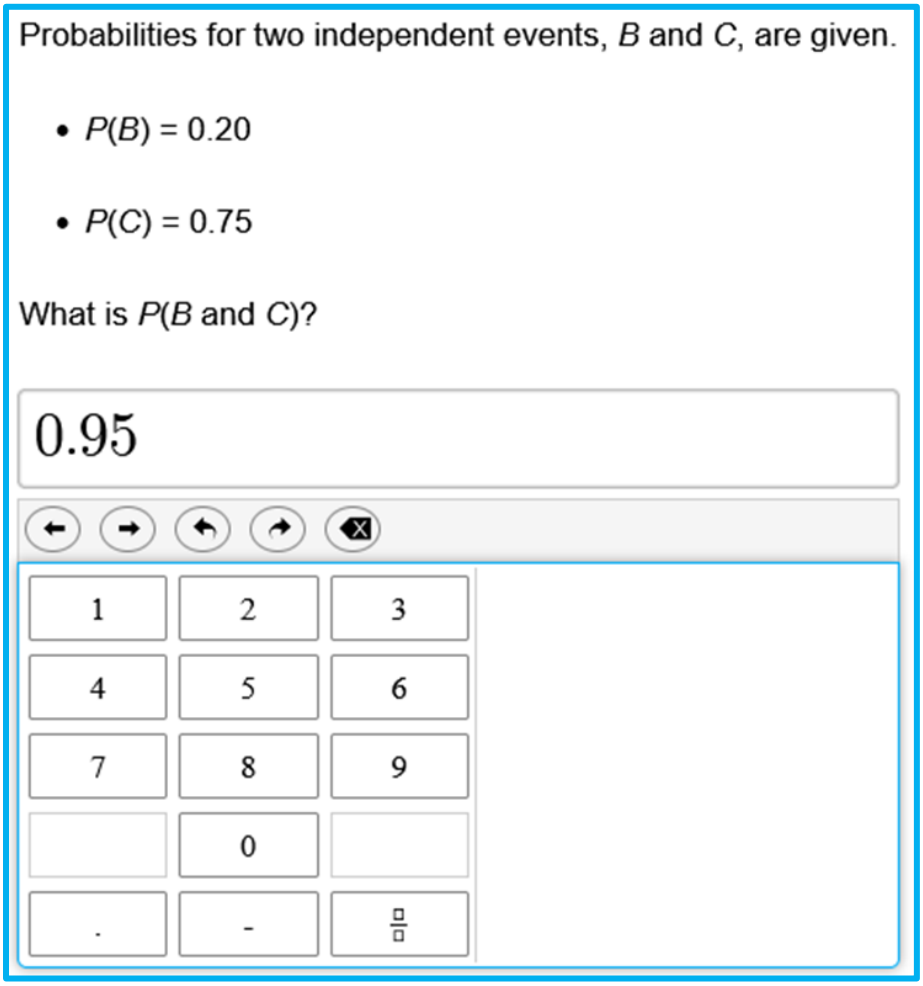
Notes on Scoring

This response receives full credit (1 point) because it provides the correct value of $P(B \text{ and } C)$.

The student understands that because the probabilities are independent, $P(B \text{ and } C) = P(B) \times P(C)$.

$$0.20 \times 0.75 = 0.15$$

Sample Response: 0 points



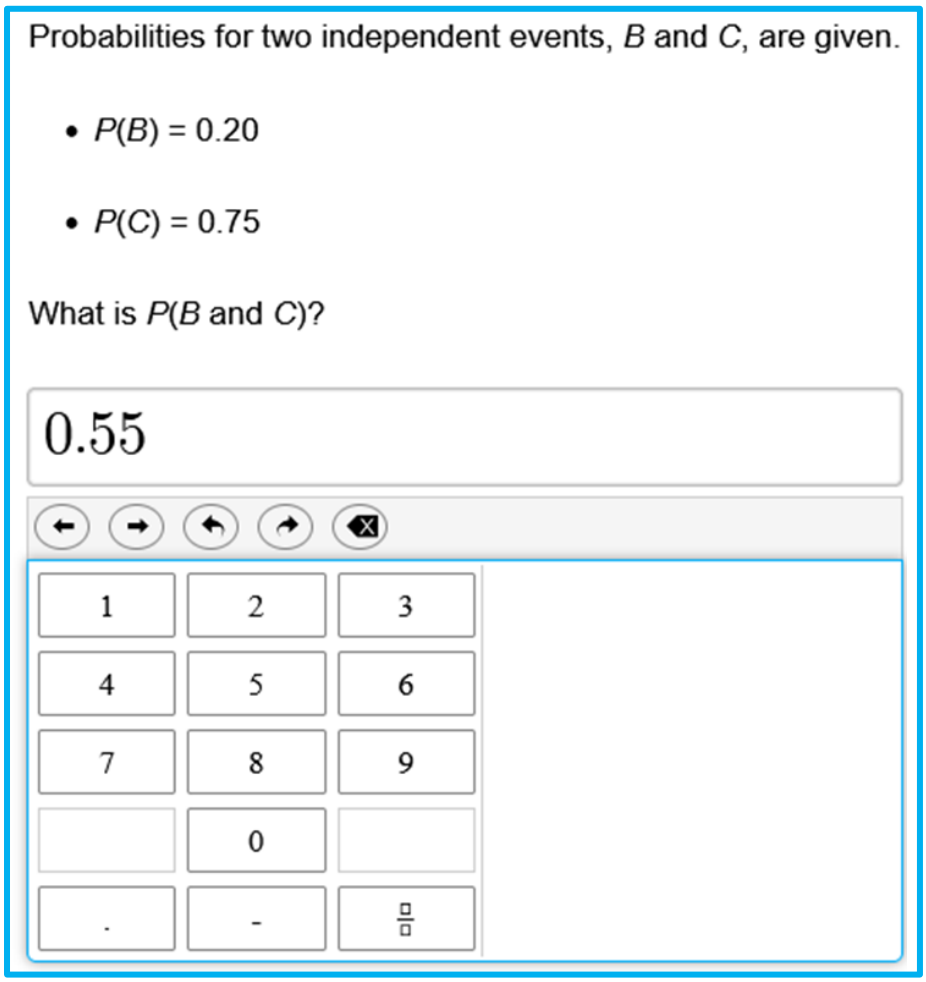
Notes on Scoring

This response receives no credit (0 points) because it does not provide the correct value of $P(B \text{ and } C)$.

The student may think that $P(B \text{ and } C)$ is calculated by adding $P(B)$ and $P(C)$ instead of multiplying.

$$0.20 + 0.75 = 0.95$$

Sample Response: 0 points



Notes on Scoring

This response receives no credit (0 points) because it does not provide the correct value of $P(B \text{ and } C)$.

The student may think that $P(B \text{ and } C)$ is calculated by subtracting $P(B) - P(C)$, instead of multiplying.

$$0.75 - 0.20 = 0.55$$

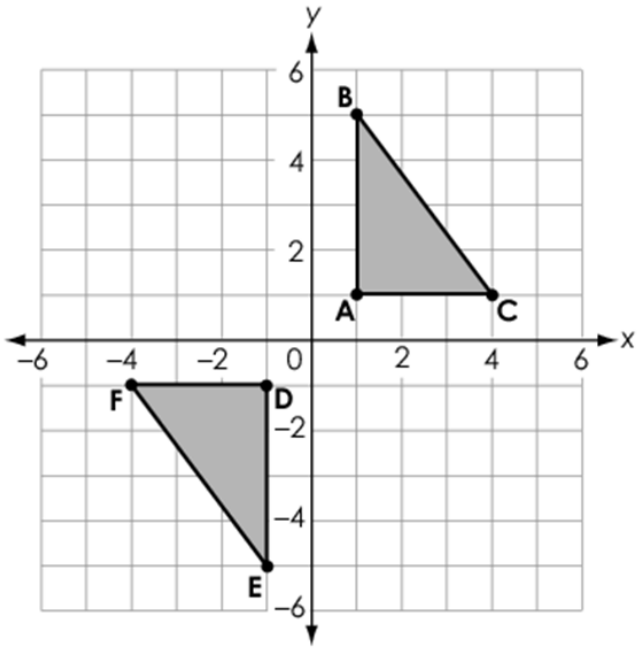
Geometry
Spring 2025 Item Release

Question 43

Question and Scoring Guidelines

Question 43

Triangle ABC is transformed to create triangle DEF as shown.



Select all of the transformations that could have been performed on triangle ABC to create triangle DEF.

- [ ] a reflection across the line $y = -x$
- [ ] a rotation of 180° clockwise about the origin
- [ ] a translation 2 units down and 2 units to the left
- [ ] a reflection across the *y*-axis, and then a reflection across the *x*-axis
- [ ] a reflection across the *x*-axis, and then a translation 2 units to the left

Points Possible: 1

Content Cluster: Experiment with transformations in the plane.

Content Standard: Given a geometric figure and a rotation, reflection, or translation, draw the transformed figure using, e.g., graph paper, tracing paper, or geometry software. Specify a sequence of transformations that will carry a given figure onto another. *(G.CO.5)*

Depth of Knowledge: Level 2

Modeling and Reasoning: No

Scoring Guidelines

Rationale for First Option: This is incorrect. The student may only consider which quadrant the resulting image ends up in and not the side lengths.

Rationale for Second Option: **Key** – The student recognizes that the coordinates of the corresponding ordered pairs are opposites, meaning that a 180-degree rotation about the origin of triangle ABC produces triangle DEF.

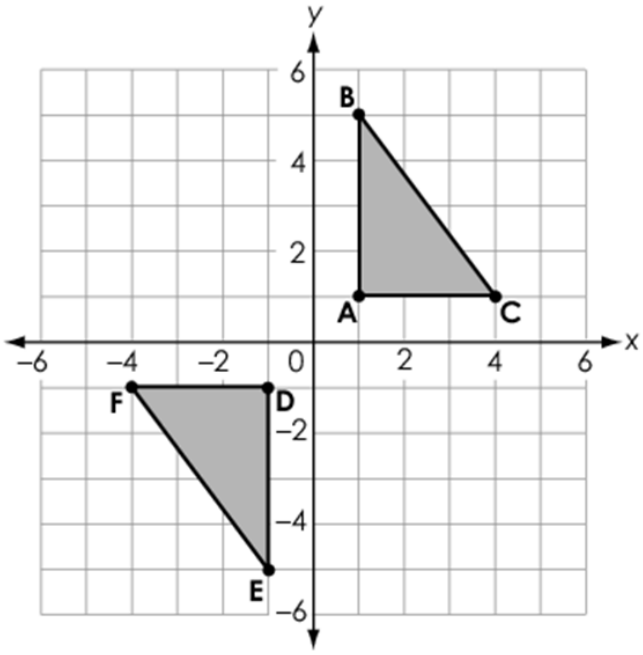
Rationale for Third Option: This is incorrect. The student may identify a translation that maps point A onto point D but does not consider the other vertices of the triangle.

Rationale for Fourth Option: **Key** – The student recognizes that each triangle is the same distance from each axis and that if triangle ABC is reflected over the *x*-axis, then the corresponding vertices of this image and triangle DEF have the same *y*-coordinates as triangle DEF. Then, since the image is the same distance from the *y*-axis as triangle DEF and the *y*-coordinates of the corresponding vertices are the same, a reflection of the image across the *y*-axis results in a new image that is triangle DEF.

Rationale for Fifth Option: This is incorrect. The student may identify a sequence of transformations that maps side AD onto side DE but does not consider the other sides of the triangle.

Sample Response: 1 point

Triangle ABC is transformed to create triangle DEF as shown.



Select all of the transformations that could have been performed on triangle ABC to create triangle DEF.

- [ ] a reflection across the line $y = -x$
- [x] a rotation of 180° clockwise about the origin
- [ ] a translation 2 units down and 2 units to the left
- [x] a reflection across the *y*-axis, and then a reflection across the *x*-axis
- [ ] a reflection across the *x*-axis, and then a translation 2 units to the left

Geometry
Spring 2025 Item Release

Question 45

Question and Scoring Guidelines

Question 45

Martin claims that for any rotation of a shape between 0 and 360 degrees about a point on the coordinate plane, there will always be one vertex of the shape that is stationary.

Which statement is true about Martin's claim?

Ⓐ Martin's claim is correct because a rotation preserves angle measures.

Ⓑ Martin's claim is correct because the center of rotation is always stationary.

Ⓒ Martin's claim is incorrect because there can be more than one vertex that is stationary after a rotation.

Ⓓ Martin's claim is incorrect because a vertex of the shape is stationary only if that vertex is the center of rotation.

Points Possible: 1

Content Cluster: Experiment with transformations in the plane.

Content Standard: Develop definitions of rotations, reflections, and translations in terms of angles, circles, perpendicular lines, parallel lines, and line segments. *(G.CO.4)*

Depth of Knowledge: Level 2

Modeling and Reasoning: Yes

Scoring Guidelines

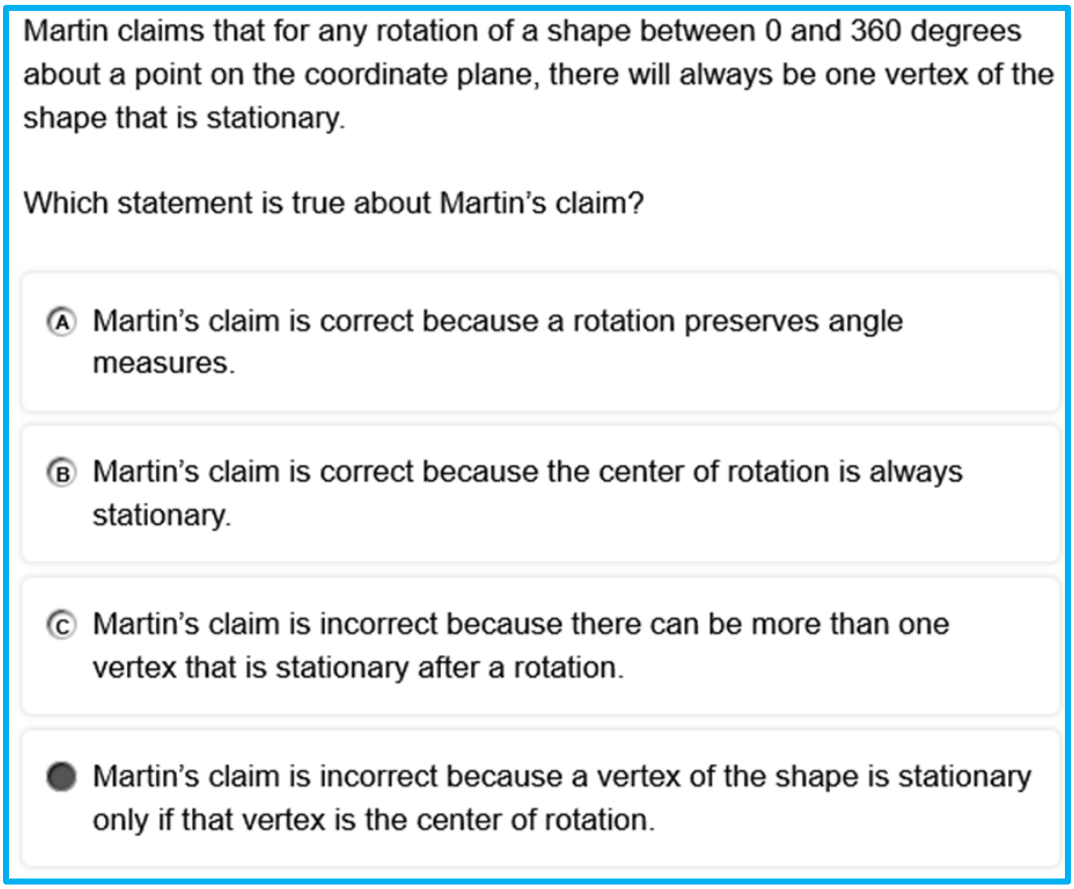
Rationale for Option A: This is incorrect. The student may confuse preservation of angle measure with preservation of location and think a vertex would not move under a rotation.

Rationale for Option B: This is incorrect. The student may realize that the center of rotation is stationary in a rotation but misses that not every rotation of a shape occurs about a vertex.

Rationale for Option C: This is incorrect. The student may think there could be two vertices that would remain stationary, such as with a reflection over a side of a shape.

Rationale for Option D: **Key** – The student identifies that shapes do not have to be rotated about a vertex; they can be rotated about a point that is not a vertex.

Sample Response: 1 point



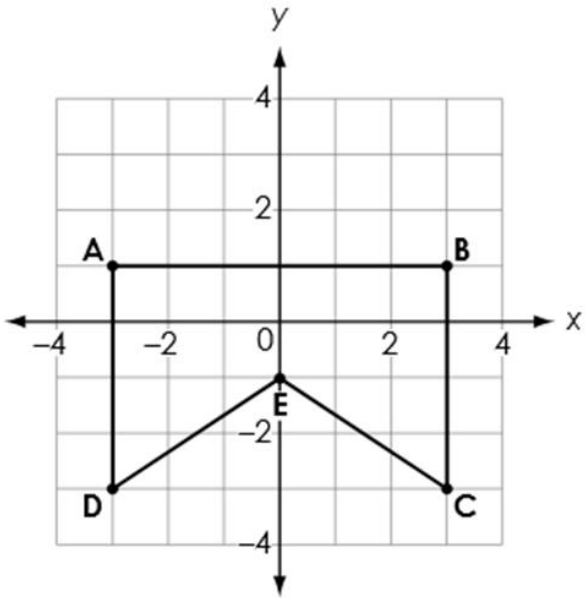
Geometry
Spring 2025 Item Release

Question 46

Question and Scoring Guidelines

Question 46

A figure is shown.



What is the area, in square units, of the figure?

square units

1 2 3
4 5 6
7 8 9
0
. -

Points Possible: 1

Content Cluster: Use coordinates to prove simple geometric theorems algebraically and to verify specific geometric statements.

Content Standard: Use coordinates to compute perimeters of polygons and areas of triangles and rectangles, e.g., using the distance formula. *(G.GPE.7)*

Depth of Knowledge: Level 2

Modeling and Reasoning: Yes

Scoring Guidelines

For full credit (1 point), the student's response satisfies the bullet below.

- The student finds the correct area, providing evidence of the ability to use coordinates to compute areas of triangles and rectangles.

Exemplar Response

- 18 square units
  or
- any equivalent value

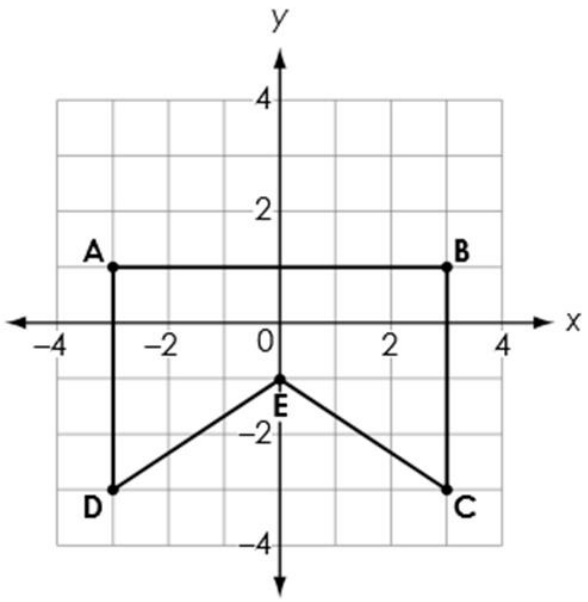
Geometry
Spring 2025 Item Release

Question 46

Sample Responses

Sample Response: 1 point

A figure is shown.



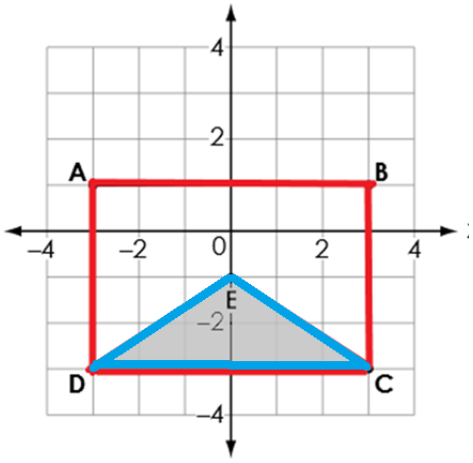
What is the area, in square units, of the figure?

18 *square units*

Notes on Scoring

This response receives full credit (1 point) because it correctly identifies the area of the figure.

The student identifies that the figure can be broken down into a **rectangle** with a **triangle** removed from it.



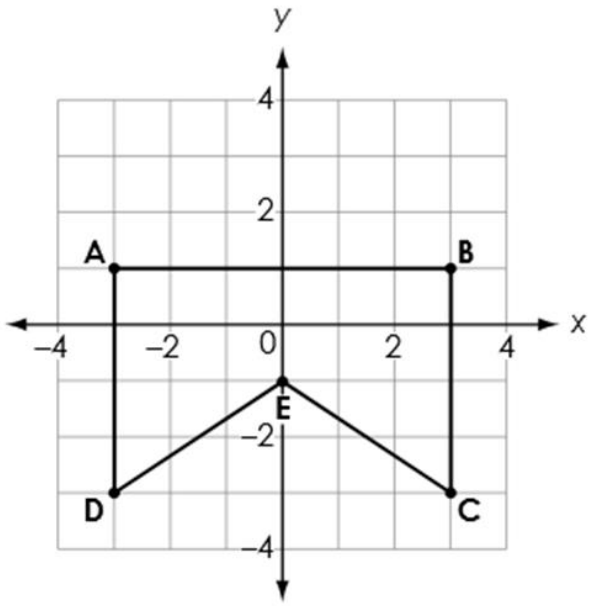
The area of the **rectangle** is 4 × 6, or 24 square units.

The area of the **triangle** is $\frac{1}{2}(2 \times 6)$, or 6 square units.

The student subtracts the area of the triangle from the area of the rectangle to calculate the area of the figure as 18 square units.

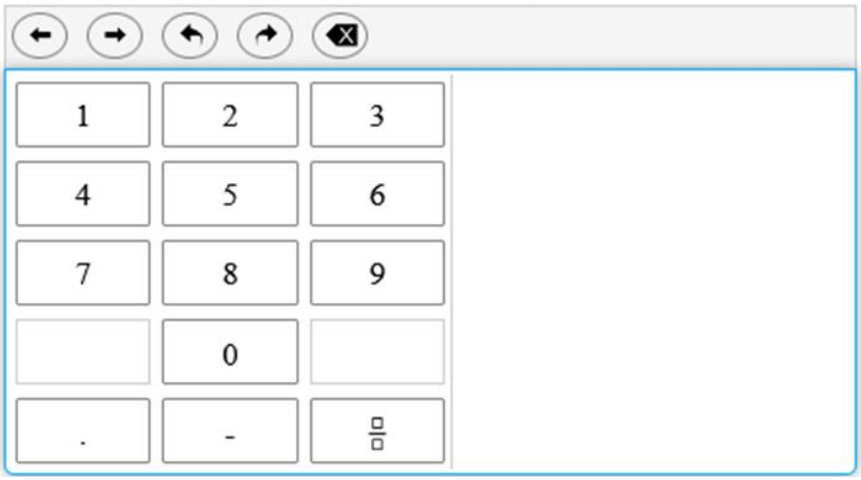
Sample Response: 0 points

A figure is shown.



What is the area, in square units, of the figure?

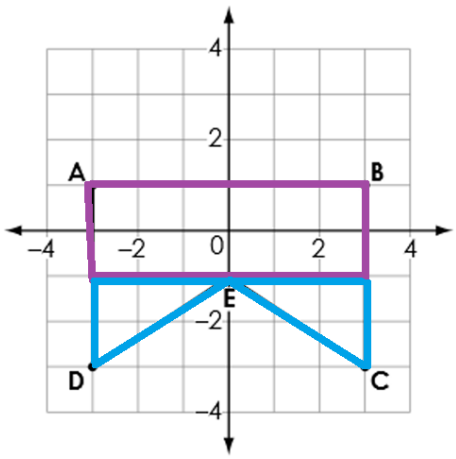
24 *square units*



Notes on Scoring

This response receives no credit (0 points) because it does not correctly identify the area of the figure.

The student may identify that the figure can be broken down into a **rectangle** and **two triangles**.



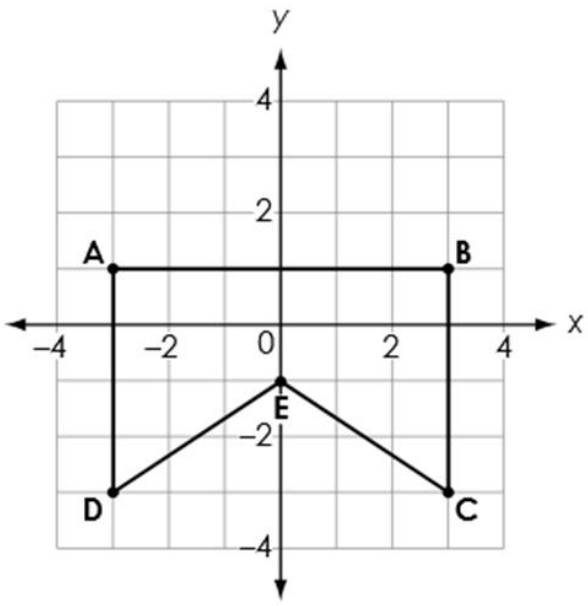
The student calculates the area of the **rectangle** is 2 × 6, or 12 square units.

The student may calculate the area of each **triangle** as (2 × 3) instead of $\frac{1}{2}(2 \times 3)$.

The student adds the area of the rectangle and the incorrect area for the two triangles to calculate an incorrect area of the figure as 12 + 6 + 6, or 24 square units.

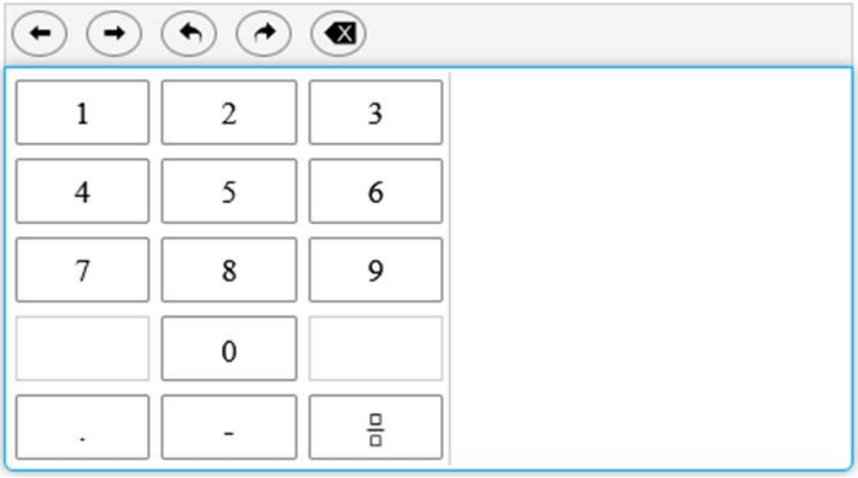
Sample Response: 0 points

A figure is shown.



What is the area, in square units, of the figure?

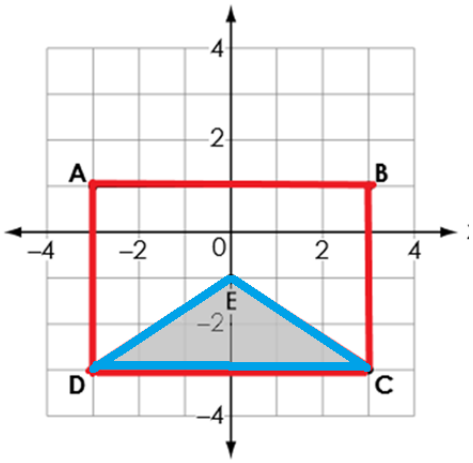
30 *square units*



Notes on Scoring

This response receives no credit (0 points) because it does not correctly identify the area of the figure.

The student identifies that the figure can be broken down into a **rectangle** with a **triangle** removed from it.



The area of the **rectangle** is 4 × 6, or 24 square units.

The area of the **triangle** is $\frac{1}{2}(2 \times 6)$, or 6 square units.

The student may add the area of the triangle to the area of the rectangle, instead of subtracting, to calculate the area of the figure as 24 + 6, or 30 square units.

Geometry
Spring 2025 Item Release

Question 51

Question and Scoring Guidelines

Question 51

Which definition most accurately describes a single angle?

Ⓐ the figure formed by two parallel rays

Ⓑ the figure formed by two intersecting rays

Ⓒ the figure formed by two rays that are not parallel

Ⓓ the figure formed by two rays with a common endpoint

Points Possible: 1

Content Cluster: Experiment with transformations in the plane.

Content Standard: Know precise definitions of ray, angle, circle, perpendicular line, parallel line, and line segment, based on the undefined notions of point, line, distance along a line, and arc length. *(G.CO.1)*

Depth of Knowledge: Level 1

Modeling and Reasoning: No

Scoring Guidelines

Rationale for Option A: This is incorrect. The student may realize that rays are involved in the definition of an angle but misses that the rays should have a common endpoint.

Rationale for Option B: This is incorrect. The student recognizes that an angle is made up of two intersecting rays but does not specify that their intersection must be at a common endpoint. Two intersecting rays can create up to four angles depending on how they are oriented.

Rationale for Option C: This is incorrect. The student recognizes that rays that are not parallel may intersect but overlooks that rays that are not parallel do not necessarily intersect, and intersecting rays do not necessarily create a single angle.

Rationale for Option D: **Key** – The student recognizes a precise definition of an angle. Two rays with a common endpoint will create one angle.

Sample Response: 1 point

